

The Itô formula for rough paths

The bracket of a rough path

Idea: the quadratic variation of a stochastic process has an important role in stochastic analysis, however a rough path (often) does not admit a quadratic variation but we can define a corresponding object: the bracket.

Bracket of a rough path: let $X = (X, \mathbb{X}) \in \mathcal{C}^\alpha$ be a rough path. The bracket of X is defined as the path $[X]: [0, T] \rightarrow \mathbb{R}^{d \times d}$ given by

$$[X]_t := X_{0,t} \otimes X_{0,t} - 2 \text{Sym}(\mathbb{X}_{0,t})$$

The bracket is a 2α -Hölder path: it holds that

$$[X]_{s,t} = X_{s,t} \otimes X_{s,t} - 2 \text{Sym}(\mathbb{X}_{s,t}) \quad \leftarrow \text{Notice that it is a symmetric object!}$$

In particular, $[X]_t \in C^{2\alpha}([0, T], \mathbb{R}^{d \times d})$.

Proof Equality:

$$\begin{aligned} [X]_{s,t} &= [X]_t - [X]_s = X_{0,t} \otimes X_{0,t} - 2 \text{Sym}(\mathbb{X}_{0,t}) - X_{0,s} \otimes X_{0,s} - 2 \text{Sym}(\mathbb{X}_{0,s}) \\ &\stackrel{\text{Sym is linear}}{=} X_{0,t} \otimes X_{0,t} - X_{0,s} \otimes X_{0,s} - 2 \text{Sym}(\mathbb{X}_{0,t} - \mathbb{X}_{0,s}) \\ &= X_{0,t} \otimes X_{0,t} - X_{0,s} \otimes X_{0,s} - 2 \text{Sym}(\mathbb{X}_{s,t} + X_{0,s} \otimes X_{s,t}) \\ &\stackrel{\text{Chen's relation}}{=} X_{0,t} \otimes X_{0,t} - X_{0,s} \otimes X_{0,s} - 2 \text{Sym}(\mathbb{X}_{s,t}) - X_{0,s} \otimes X_{s,t} - X_{s,t} \otimes X_{0,s} \\ &\stackrel{\text{Sym linear and } \text{Sym}(a \otimes b) = a \otimes b + b \otimes a}{=} \cancel{X_0 \otimes X_0} - \cancel{X_0 \otimes X_t} - \cancel{X_t \otimes X_0} + X_t \otimes X_t - \cancel{X_0 \otimes X_s} - \cancel{X_s \otimes X_0} + \cancel{X_s \otimes X_s} \\ &\quad - 2 \text{Sym}(\mathbb{X}_{s,t}) - \cancel{X_0 \otimes X_s} + \cancel{X_0 \otimes X_t} + X_s \otimes X_s - X_s \otimes X_t - \cancel{X_s \otimes X_0} + \cancel{X_s \otimes X_s} \\ &\quad + \cancel{X_t \otimes X_0} - X_t \otimes X_s \\ &= X_{s,t} \otimes X_{s,t} - 2 \text{Sym}(\mathbb{X}_{s,t}). \end{aligned}$$

Regularity:

$$|[X]_{s,t}| \leq |X_{s,t} \otimes X_{s,t}| + |2 \text{Sym}(\mathbb{X}_{s,t})| \leq \|X\|_\alpha |t-s|^{2\alpha} + \|\mathbb{X}_{s,t}\|_{2\alpha} |t-s|^{2\alpha}$$

Weakly geometric rough path: $X \in \mathcal{C}^\alpha$ is weakly geometric $\Leftrightarrow [X]_t = 0 \quad \forall t \in [0, T]$.

and bracket

(straightforward from definition)

Bracket of $B^{\text{Itô}}$: let $B^{\text{Itô}} = (B, \bar{B}^{\text{Itô}})$ the Itô rough path. Then

$$[B^{\text{Itô}}]_{s,t} = (t-s)I$$

↓

$$\text{Proof: } \text{Sym}(B^{\text{Itô}}_{s,t}) = \frac{1}{2} (B_{s,t} \otimes B_{s,t} - (t-s)I)$$

Remark: in the previous prop. we have shown that, at least for the Itô Brownian rough path, the bracket coincide with the quadratic variation of the brownian motion as stochastic process.

The bracket of a rough integral: $X = (X, \bar{X}) \in \mathcal{C}^\alpha([0, T], \mathbb{R}^d)$ and

$(K, K') \in \mathcal{D}_X^{2\alpha}([0, T], \mathcal{L}(\mathbb{R}^d, \mathbb{R}^m))$. Recall that $(Z, Z') := (\int_0^\cdot K_u d\bar{X}_u, K_u) \in \mathcal{D}_X^{2\alpha}([0, T], \mathbb{R}^m)$

and that we can define $Z := (Z, \mathbb{Z}) \in \mathcal{C}^\alpha([0, T]; \mathbb{R}^m)$ with $\mathbb{Z}$ the canonical lifting for controlled path. Then

$$[Z]_t = \int_0^t (K_u \otimes K_u) d[X]_u \in \mathbb{R}^m \times \mathbb{R}^m$$

this is a Young Integral! In fact $[X] \in \mathcal{C}^{2\alpha}$ and $K \otimes K \in \mathcal{C}^\alpha$
 $\Rightarrow 2\alpha + \alpha > 1$. In particular this integral is:

$$\lim_{|\pi| \rightarrow 0} \sum_{[u,v] \in \pi} (K_u \otimes K_u) [X]_{u,v} \in \mathbb{R}^m \otimes \mathbb{R}^m = \mathbb{R}^m \times \mathbb{R}^m$$

$\mathcal{L}(\mathbb{R}^d, \mathbb{R}^m) \otimes \mathcal{L}(\mathbb{R}^d, \mathbb{R}^m) \overset{\cap}{=} \mathbb{R}^d \times \mathbb{R}^d = \mathbb{R}^d \otimes \mathbb{R}^d$
 $\simeq \mathcal{L}(\mathbb{R}^d \otimes \mathbb{R}^d, \mathbb{R}^m \otimes \mathbb{R}^m)$

→ Proof: $[X]_t = Z_{s,t} \otimes Z_{s,t} - 2 \text{Sym}(\mathbb{Z}_{s,t}) \overset{\text{by the bound of the rough integral}}{=} Z_{s,t} = K_s X_{s,t} + K'_s \bar{X}_{s,t} + O(|t-s|^{3\alpha})$

$(K'_s \bar{X}_{s,t})^{\otimes 2}$ has 4α -order,

$$K'_s \bar{X}_{s,t} \otimes K_s X_{s,t} \overset{\wedge}{=} (K_s X_{s,t} + K'_s \bar{X}_{s,t} + O(|t-s|^{3\alpha})) \otimes (K_s X_{s,t} + K'_s \bar{X}_{s,t} + O(|t-s|^{3\alpha})) - 2 \text{Sym}(\mathbb{Z}_{s,t})$$

and $K_s X_{s,t} \otimes K'_s \bar{X}_{s,t}$

have 3α -order

$$= (K_s X_{s,t}) \otimes (K_s X_{s,t}) - 2 \text{Sym}(\mathbb{Z}_{s,t}) + O(|t-s|^{3\alpha}) = (*)$$

Now we focus on $\text{Sym}(\mathbb{Z}_{s,t})$:

$$Z_{s,t} := \int_s^t Z_{s,r} \otimes dZ_r = \underbrace{Z_s \otimes Z_{s,t} + (Z'_s \otimes Z'_s) X_{s,t}}_{\text{bound on the integral of controlled path against another one}} + O(|t-s|^{3\alpha})$$

$$Z' = K$$

$$\rightarrow = Z_s \otimes Z_{s,t} + (K_s \otimes K_s) X_{s,t} + O(|t-s|^{3\alpha})$$

At some point it gets rid off $\rightarrow$

$$Z_s \otimes Z_{s,t}$$



$$\text{Sym}(Z_{s,t}) = \text{Sym}((K_s \otimes K_s) X_{s,t}) + O(|t-s|^{3\alpha})$$

$$K_s \otimes K_s \text{ is sym} \rightarrow = (K'_s \otimes K_s) \text{Sym}(X_{s,t}) + O(|t-s|^{3\alpha})$$

$$\text{So: } (*) = (K_s X_{s,t}) \otimes (K_s X_{s,t}) - 2 (K'_s \otimes K_s) \text{Sym}(X_{s,t}) + O(|t-s|^{3\alpha})$$

$$= (K_s \otimes K_s) (X_{s,t} \otimes X_{s,t}) - 2 (K'_s \otimes K_s) \text{Sym}(X_{s,t}) + O(|t-s|^{3\alpha})$$

$$= (K_s \otimes K_s) (X_{s,t} \otimes X_{s,t} - 2 \text{Sym}(X_{s,t})) + O(|t-s|^{3\alpha})$$

$$= (K_s \otimes K_s) [\bar{X}]_{s,t} + O(|t-s|^{3\alpha})$$

Taking $\lim_{|t-s| \rightarrow 0} \sum_{[s,t] \in \mathcal{T}}$ on both sides we have the result. $\square$

Remark: with an abuse of notation we have shown $\left[\int_0^\cdot K_u d\bar{X}_u \right]_t = \int_0^t K_u^z d[\bar{X}]_u$

that remind us the Itô-isometry formula.

The Itô formula for rough paths

Itô formula for rough path: let $\bar{X} = (X, X') \in \mathcal{C}^\alpha([0, T], \mathbb{R}^d)$ and let $f \in C^3(\mathbb{R}^d; \mathbb{R}^m)$. Then

$$f(X_T) = f(X_0) + \int_0^T Df(X_u) d\bar{X}_u + \frac{1}{2} \int_0^T D^2 f(X_u) d[\bar{X}]_u$$

this is the rough integral
of $(Df(X), D^2 f(X))$
against $\bar{X}$

this is the Young integral
of $D^2 f X$ against $[\bar{X}]$

notice that since $f \in C^3 \Rightarrow Df \in C^2$

$$\Rightarrow (Df, D^2 f) \in \mathcal{V}_X^{2\alpha}([0, T]; \mathcal{L}(\mathbb{R}^d, \mathbb{R}^m))$$

and so the rough integral is well-defined

as the Young integral $(D^2 f \in C^\alpha, [\bar{X}] \in C^{2\alpha})$

→ Remark (Taylor expansions): given $f: \mathbb{R}^d \rightarrow \mathbb{R}^m$ with $f \in C^{n+1}$, then $\forall x, h \in \mathbb{R}^d$

$$f(x+h) = \sum_{k=0}^n \frac{1}{k!} D^k f(x) (h^{\otimes k}) + \underbrace{\int_{[0,1]^{n+1}} D^{n+1}(f(x+r_1-r_{n+1}h)) (h^{\otimes n+1}) r_1^n r_2^{n-1} \dots r_n dr_{n+1} \dots dr_1}_{= \int_{[0,1]^n} (D^n f(x+r_1-r_n h) - D^n f(x)) (h^{\otimes n}) r_1^{n-1} r_2^{n-2} \dots r_{n-1} dr_n \dots dr_1}$$

$$\begin{aligned} & (D^n f(x+r_1-r_n h) - D^n f(x)) (h^{\otimes n}) \\ &= \int_0^1 D^{n+1} f(x+r_1-r_{n+1}h) (h^{\otimes n+1}) r_1-r_n dr_{n+1} \end{aligned}$$

→ Remark (Contraction of symm. against ant.): if you have A_{ij} symmetric and B^{ij} antisymmetric:

$$\sum_{i,j} A_{ij} B^{ij} = - \sum_{i,j} A_{ji} B^{ji} = - \sum_{i,j} A_{ij} B^{ij} \Rightarrow \sum_{i,j} A_{ij} B^{ij} = 0$$

$\begin{matrix} \uparrow \\ B \text{ anty.} \\ A \text{ symm.} \end{matrix}$
 $\begin{matrix} \uparrow \\ \text{do the sum on } i,j \\ \text{it's the same on } j,i \end{matrix}$

→ Proof: we use the Taylor expansion for $n=2$, $x = X_s$, $h = X_{s,t}$ so $x+h = X_t$

$$f(X_t) = f(X_s) + Df(X_s)(X_{s,t}) + \frac{1}{2} D^2 f(X_s)(X_{s,t} \otimes X_{s,t}) + \underbrace{\int_0^1 \int_0^1 (D^2 f(X_s + r_1 r_2 X_{s,t}) - D^2 f(X_s))(X_{s,t} \otimes X_{s,t}) r_1 dr_2 dr_1}_{R_{s,t}}$$

Now let us analyze the term $\frac{1}{2} D^2 f(X_s)(X_{s,t} \otimes X_{s,t})$:

$$\frac{1}{2} D^2 f(X_s)(X_{s,t} \otimes X_{s,t}) = \frac{1}{2} D^2 f(X_s)([X]_{s,t} + 2 \text{Sym}(X_{s,t})) =$$

$$D^2 f(X_s)(X_{s,t}) = \frac{1}{2} D^2 f(X_s)[X]_{s,t} + D^2 f(X_s)(\text{Sym}(X_{s,t}))$$

$$= D^2 f(X_s)(\text{Sym}(X_{s,t})) + \frac{1}{2} D^2 f(X_s)[X]_{s,t} + D^2 f(X_s)(X_{s,t})$$

$$D^2 f(X_s)(\text{Ant}(X_{s,t})) \stackrel{\uparrow}{=} D^2 f(X_s)(\text{Sym}(X_{s,t}))$$

$D^2 f(X_s)$ is symmetric (the Hessian)

so use the remark

Concerning the remainder $R_{s,t}$:

$$|R_{s,t}| \stackrel{\uparrow}{\leq} \|f\|_{C_b^3} |X_{s,t}|^3 \leq \|f\|_{C_b^3} \|X\|_\alpha^3 |t-s|^{3\alpha} = O(|t-s|^{3\alpha})$$

as always, $X([0,T])$ is compact and

f is continuous so $f(X)$ is bounded, maybe is more useful to use the 1st scripture of the Taylor expansion

So we can rewrite the Taylor expansion as:

$$f(X_t) - f(X_s) = \underbrace{Df(X_s)X_{s,t} + D^2 f(X_s)(X_{s,t})}_{\leftarrow} + \underbrace{\frac{1}{2} D^2 f(X_s)[X]_{s,t}}_{\leftarrow} + O(|t-s|^{3\alpha})$$

taking $\lim_{|t-s| \rightarrow 0} \sum_{[s,t] \in \Pi}$ the LHS is simply $f(X_T) - f(X_0)$, this becomes the rough integral, this

becomes the Young integral and $O(|t-s|^{3\alpha}) \rightarrow 0$ since $3\alpha > 1$. $\square$

Ito formula for a special case of controlled path: let $\bar{X} = (X, X) \in C^\alpha([0, T]; \mathbb{R}^d)$ and suppose

$$(Y, Y') \in D_X^{2\alpha}([0, T], \mathbb{R}^m), \quad (Y', Y'') \in D_X^{2\alpha}([0, T], \mathcal{Z}(\mathbb{R}^d, \mathbb{R}^m)) \quad \text{s.t.} \quad \exists \Gamma \in C^\alpha([0, T], \mathbb{R}^m):$$

$$Y_t = Y_0 + \int_0^t Y'_s d\bar{X}_s + \Gamma_t \quad \forall t \in [0, T].$$

Then, $\forall t \in C^3$ we have:

$$f(Y_T) = f(Y_0) + \int_0^T Df(Y_u) Y'_u d\bar{X}_u + \int_0^T Df(Y_u) d\Gamma_u + \frac{1}{2} \int_0^T D^2 f(Y_u) (Y'_u \otimes Y'_u) d[\bar{X}]_u$$

→ Proof: we proceed as before, from the Taylor expansion since $Y \in C^\alpha$ we know:

$$f(Y_t) - f(Y_s) = Df(Y_s) Y_{s,t} + \frac{1}{2} D^2 f(Y_s) (Y_{s,t} \otimes Y_{s,t}) + O(|t-s|^{3\alpha})$$

Now using the bound on the rough integral:

$$Y_t = \underset{\uparrow \text{hp.}}{Y_s} + \int_s^t Y'_s d\bar{X}_s + \Gamma_{s,t} = Y_s + Y'_s X_{s,t} + Y''_s X_{s,t} + \Gamma_{s,t} + O(|t-s|^{3\alpha})$$

So substituting in the Taylor expansion:

all the other terms have order $\geq 3\alpha$

↓

$$f(Y_t) - f(Y_s) = Df(Y_s) (Y'_s X_{s,t} + Y''_s X_{s,t} + \Gamma_{s,t}) + \frac{1}{2} D^2 f(Y_s) (Y'_s X_{s,t} \otimes Y'_s X_{s,t}) + O(|t-s|^{3\alpha})$$

$$\text{as in the previous} \rightarrow = Df(Y_s) Y'_s X_{s,t} + Df(Y_s) Y''_s X_{s,t} + Df(Y_s) \Gamma_{s,t}$$

proof we know

$$+ \frac{1}{2} D^2 f(Y_s) (Y'_s \otimes Y'_s) (X_{s,t} \otimes X_{s,t}) + D^2 f(Y'_s \otimes Y'_s) (X_{s,t})$$

$$D^2 f(Y'_s \otimes Y'_s) (X_{s,t}) =$$

$$= D^2 f(Y'_s \otimes Y'_s) (\text{Sym } X_{s,t})$$

$$- D^2 f(Y'_s \otimes Y'_s) (\text{Sgm } X_{s,t}) + O(|t-s|^{3\alpha})$$

so: add and subtract

$$\begin{aligned} &= Df(Y_s) Y'_s X_{s,t} + \overbrace{(Df(Y_s) Y''_s + D^2 f(Y'_s \otimes Y'_s))}^{(*)} X_{s,t} + Df(Y_s) \Gamma_{s,t} \\ &\quad + \frac{1}{2} D^2 f(Y_s) (Y'_s \otimes Y'_s) \left(\underbrace{X_{s,t} \otimes X_{s,t} - 2 \text{Sgm } X_{s,t}}_{= [\bar{X}]_{s,t}} \right) + O(|t-s|^{3\alpha}) \\ &= [\bar{X}]_{s,t} \end{aligned}$$

Let us focus on $(*)$: by the product rule

$$\begin{aligned} (Df(Y) Y')' &= (Df(Y))' Y' + Df(Y) (Y')' = \leftarrow \text{the Gubinelli derivative of } Df(Y) \text{ when } \\ &\quad (Y, Y') \in D_X^{2\alpha} \text{ is } D^2 f(Y) (Y' \otimes Y') \text{ instead by} \\ &= D^2 f(Y) Y' Y' + Df(Y) Y'' \end{aligned}$$

✓ see first lemma on "Functions of controlled path" section

hip. the Gubinelli derivative of Y' is Y''

$$\text{so } (*) = (Df(Y_s) Y'_s)'$$

So we have

$$f(Y_t) - f(Y_s) = Df(Y_s) Y'_s X_{s,t} + (Df(Y_s) Y'_s)' X_{s,t} + Df(Y_s) \Gamma_{s,t} \\ + \frac{1}{2} D^2 f(Y_s) (Y'_s \otimes Y'_s) [\mathbf{X}]_{s,t} + O(|t-s|^{3\alpha})$$

Taking the $\lim_{|\pi| \rightarrow 0} \sum_{[s,t] \in \pi}$ we obtain the result.

□

The rough exponential

Notation: we write $\|X\|_{\alpha, [s,t]}$ for the α -seminorm of the interval $[s,t]$; $\|\mathbf{X}\|_{2\alpha, [s,t]}$ and

$\|\mathbf{X}\|_{\alpha, [s,t]}$ similarly

We see our first example of rough differential equation (RDE).

The rough exponential: let $\alpha \in (\frac{1}{3}, \frac{1}{2}]$ and $\mathbf{X} = (X, \mathbf{X}) \in \mathcal{C}^{\beta}([0, T], \mathbb{R})^{\mathbb{R}!}$ s.t. $X_0 = 0$.

Let

$$V_t = \exp\left(X_t - \frac{1}{2} [\mathbf{X}]_t\right), \quad t \in [0, T] \quad (\text{rough exponential}).$$

Then, V is the unique solution to the linear rough differential equation

$$V_t = 1 + \int_0^t V_u d\mathbf{X}_u$$

(by a solution we mean a path $V \in \mathcal{C}^{\beta}([0, T], \mathbb{R})$ s.t. $(V, V) \in \mathcal{D}_X^{2\beta}$ and the equation holds with the integral seen as the rough integral of (V, V) against $\mathbf{X}$).

Proof: Satisfy the RDE: we define $Y_t := X_t - \frac{1}{2} [\mathbf{X}]_t$ and $Y'_t = 1$. Then

$$(Y, Y') \in \mathcal{D}_X^{2\beta} \quad (R_{s,t} = Y_{s,t} - Y'_s X_{s,t} = X_{s,t} + \frac{1}{2} [\mathbf{X}]_{s,t} - X_{s,t} = \frac{1}{2} [\mathbf{X}]_{s,t} \in \mathcal{C}^{2\beta})$$

$$\text{and } (Y', Y''=0) \in \mathcal{D}_X^{2\beta} \text{ (obviously) and } Y_0 + \int_0^t Y'_s d\mathbf{X}_s + \Gamma_t = 0 + X_t + \Gamma_t$$

so choosing $\Gamma_t := -\frac{1}{2} [\mathbf{X}]_t$ we are under the lip
 (it's $(1,0)$ against $(X, \mathbf{X})$)
 (it's the Young Integral)

of the Itô formula for controlled paths, then choosing $f = \exp(\cdot)$ we have

$$V_t = 1 + \int_0^t V_u d\mathbf{X}_u - \frac{1}{2} \int_0^t V_u d[\mathbf{X}]_u + \frac{1}{2} \int_0^t V_u [\mathbf{X}]_u = 1 + \int_0^t V_u d\mathbf{X}_u$$

Uniqueness: let us take $\alpha \in (\frac{1}{3}, \beta)$. Since $\alpha < \beta$, it is clear that if $(\tilde{V}, \tilde{V}) \in \mathcal{D}_X^{2\beta}$ is

another solution, then $\mathbf{X} \in \mathcal{C}^{\alpha}$, $(V, \tilde{V}) \in \mathcal{D}_X^{2\alpha}$ and $(\tilde{V}, \tilde{V}) \in \mathcal{D}_X^{2\alpha}$.

By the prop. about the continuity of the rough integral we know:

$$\begin{aligned} \|V - \tilde{V}\|_\alpha &\stackrel{\text{they're solution}}{\leq} \left\| \int_0^\cdot V_u d\mathbf{X}_u - \int_0^\cdot \tilde{V}_u d\mathbf{X}_u \right\|_\alpha \stackrel{\text{the rough path it's the same and } V' = V \text{ and } V_0 = \tilde{V}_0}{\leq} C \left(1 + 2 \|\mathbf{X}\|_\alpha \right) \left(0 + 0 + (0 + \|V - \tilde{V}\|_\alpha + \|R^V - R^{\tilde{V}}\|_{2\alpha}) \|\mathbf{X}\|_\alpha \right) \\ &\leq \tilde{C} \left(\|V - \tilde{V}\|_\alpha + \|R^V - R^{\tilde{V}}\|_{2\alpha} \right) \|\mathbf{X}\|_\alpha \end{aligned}$$

By the boundary on $R^V - R^{\tilde{V}}$:

$$\begin{aligned} \|R^V - R^{\tilde{V}}\|_{2\alpha} &= \|R \int_0^\cdot V_u d\mathbf{X}_u - R \int_0^\cdot \tilde{V}_u d\mathbf{X}_u\|_{2\alpha} \\ &\leq C \left((0 + \|V - \tilde{V}\|_\alpha + \|R^V - R^{\tilde{V}}\|_{2\alpha}) \|\mathbf{X}\|_\alpha + 0 \right) \end{aligned}$$

So taking the max between $\tilde{C}$ and C (denote $\bar{C}$):

$$\underbrace{\|V - \tilde{V}\|_\alpha + \|R^V - R^{\tilde{V}}\|_{2\alpha}} \leq 2 \bar{C} \|\mathbf{X}\|_\alpha \underbrace{\left(\|V - \tilde{V}\|_\alpha + \|R^V - R^{\tilde{V}}\|_{2\alpha} \right)} \quad (*)$$

they are the same, so if we can find a way to see that $2 \bar{C} \|\mathbf{X}\|_\alpha < 1$ we must have $\|V - \tilde{V}\|_\alpha = 0$ and since the starting point it's the same we would have the result.

Now:

$$\begin{aligned} \|\mathbf{X}\|_{\alpha, [0, t]} &= \|X\|_{\alpha, [0, t]} + \|\mathbf{X}\|_{2\alpha, [0, t]} \leq \|X\|_{\beta, [0, t]} t^{\beta-\alpha} + \|\mathbf{X}\|_{2\beta, [0, t]} t^{2(\beta-\alpha)} \\ &\stackrel{\sup_{u \leq v \in I} \frac{|X|}{|u-v|^\alpha} = \sup_{u \leq v \in I} \frac{|X|}{|u-v|^\beta}}{\leq} \|X\|_{\beta, [0, T]} t^{\beta-\alpha} + \|\mathbf{X}\|_{2\beta, [0, T]} t^{2(\beta-\alpha)} \end{aligned}$$

Hence we can find t_0 s.t. $2\bar{C}\|\mathbf{X}\|_{\alpha, [0, t_0]} < 1$ and so rewriting (*)

with $\|\cdot\|_{\alpha, [0, t_0]}$ we have the result on $[0, t_0]$ (obviously (*) hold since we can think $[0, T]$ as $[0, t_0]$). Then, we can do the same procedure starting at t_0

(since $V_{t_0} = \tilde{V}_{t_0}$): $\bar{C}$ does not depend on the starting point, setting $\tilde{X}_u := X_{t_0+u} - X_{t_0}$

$\tilde{X}_{u,v} := X_{t_0+u, t_0+v}$ we obtain the solution on $[0, t_0]$ for the equation

$$\tilde{V}_u = V_{t_0} + \int_0^u \tilde{V}_r d\mathbf{X}_r$$

the estimates are the same since

$$\|\tilde{X}\|_{\beta, [0, t]} = \|X\|_{\beta, [t_0, t_0+t]} \leq \|X\|_{\beta, [0, T]} \text{ same for } \tilde{\mathbf{X}}.$$

So we want to see that $V_t := \tilde{V}_{t-t_0}$ satisfy the initial eq. for $t \in [t_0, 2t_0]$:

$$V_{t_0} = \tilde{V}_{t-t_0} = V_{t_0} + \int_0^{t-t_0} \tilde{V}_r d\tilde{X}_r = \left(1 + \int_0^{t_0} V_r d\tilde{X}_r \right) + \int_0^{t-t_0} \tilde{V}_r d\tilde{X}_r$$

$$\leftarrow = 1 + \int_0^{t_0} V_r d\tilde{X}_r + \int_{t_0}^t V_r d\tilde{X}_r$$

invariance of
temporal translation:

$$\int_{t_0}^{t_0+u} V_r d\tilde{X}_r =$$

$$= \int_0^u \tilde{V}_s d\tilde{X}_s$$

by def. of rough int.

□

Corollary: $\tilde{X} = (X, \mathbb{X}) \in \mathcal{C}^q$ and $(K, K') \in \mathcal{D}_X^{2q}$ are s.t. $\int_0^\cdot K_u d\tilde{X}_u$ takes value in $\mathbb{R}$.

Let V be the path given by:

$$V_t = \exp \left(\int_0^t K_u d\tilde{X}_u + \frac{1}{2} \int_0^t (K_u \otimes K_u) d[\tilde{X}]_u \right) \quad t \in [0, T]$$

Prove that V is the unique solution of the RDE:

$$V_t = 1 + \int_0^t V_u K_u d\tilde{X}_u \quad t \in [0, T]$$

→ Proof: we want to rewrite the problem in terms of an exponential rough path.

So we define $(Z, Z') := \left(\int_0^\cdot K_u d\tilde{X}_u, K \right) \in \mathcal{D}_X^{2q}$ and then $Z := (Z, \mathbb{Z})$

with $\mathbb{Z}$ the canonical lift for controlled path.

$$\text{Now } 1 + \int_0^t V_u K_u d\tilde{X}_u \underset{\substack{\uparrow \\ \text{associativity}}}{=} 1 + \int_0^t V_u dZ_u \underset{\substack{\uparrow \\ \text{consistency}}}{=} 1 + \int_0^t V_u dZ_u.$$

So V satisfy our RSE $\Leftrightarrow V$ is the rough exponential (w.r.t. Z). We know

that it is unique and it's given by

$$V_t = \exp \left(Z_t - \frac{1}{2} [\mathbb{Z}]_t \right) \underset{\substack{\uparrow \\ \text{previous prop.}}}{=} \exp \left(\int_0^t K_u d\tilde{X}_u - \frac{1}{2} \int_0^t (K_u \otimes K_u) d[\tilde{X}]_u \right)$$

$$[\mathbb{Z}]_t = \int_0^t (K_u \otimes K_u) d[\tilde{X}]_u$$

□

Differential equations

Notation: throughout this section we will use $\lesssim$ to mean an inequality up to a multiplicative constant which may depend on α, T and $\|f\|_{C_b^K}$ for $K=2$ or 3 .

Young differential equations

Lemma: let $\alpha \in (0, 1]$ and $f \in C_b^2$. Then there exists a constant C , depending only on α, T and $\|f\|_{C_b^2}$ s.t. $\forall \gamma, \tilde{\gamma} \in C^\alpha$:

$$\|f(\gamma) - f(\tilde{\gamma})\|_\alpha \leq C (1 + \|\gamma\|_\alpha + \|\tilde{\gamma}\|_\alpha) (|\gamma_0 - \tilde{\gamma}_0| + \|\gamma - \tilde{\gamma}\|_\alpha).$$

Proof: $|f(\gamma)_{s,t} - f(\tilde{\gamma})_{s,t}| = |(f(\gamma_t) - f(\tilde{\gamma}_t)) + (f(\gamma_s) - f(\tilde{\gamma}_s))| =$

$$\begin{aligned} \phi(r) &:= f(\tilde{\gamma}_t + r(\gamma_t - \tilde{\gamma}_t)) \stackrel{\text{f}}{=} \int_0^1 Df(\tilde{\gamma}_t + r(\gamma_t - \tilde{\gamma}_t)) (\gamma_t - \tilde{\gamma}_t) dr - \int_0^1 Df(\tilde{\gamma}_s + r(\gamma_s - \tilde{\gamma}_s)) (\gamma_s - \tilde{\gamma}_s) dr \\ f(\gamma_t) - f(\tilde{\gamma}_t) &= \phi(1) - \phi(0) = \int_0^1 \phi'(r) dr \\ \text{add and subtract} \quad &\rightarrow = \left| \int_0^1 Df(\tilde{\gamma}_t + r(\gamma_t - \tilde{\gamma}_t)) (\gamma_{s,t} - \tilde{\gamma}_{s,t}) + \right. \\ &\quad \left. \int_0^1 Df(\tilde{\gamma}_t + r(\gamma_t - \tilde{\gamma}_t)) (\gamma_s - \tilde{\gamma}_s) \right. \\ &\quad \left. + \int_0^1 [Df(\tilde{\gamma}_t + r(\gamma_t - \tilde{\gamma}_t)) - Df(\tilde{\gamma}_s + r(\gamma_s - \tilde{\gamma}_s))] (\gamma_s - \tilde{\gamma}_s) dr \right| \end{aligned}$$

1st term:

$$|Df(\cdot)(a-b)| \leq \|Df\|_{C_b^2} |a-b| \lesssim |\gamma_{s,t} - \tilde{\gamma}_{s,t}| + \int_0^1 \underbrace{|\tilde{\gamma}_t + r(\gamma_t - \tilde{\gamma}_t) - \tilde{\gamma}_s - r(\gamma_s - \tilde{\gamma}_s)|}_{= |\tilde{\gamma}_{s,t} + r(\gamma_{s,t} - \tilde{\gamma}_{s,t})|} |\gamma_s - \tilde{\gamma}_s| dr$$

2nd term:

$$|Df(v) - Df(w)| \leq \|Df\|_{C_b^2} |v-w| \leq |\tilde{\gamma}_{s,t}| + 2|\gamma_{s,t}| \lesssim |\tilde{\gamma}_{s,t}| + |\gamma_{s,t}|$$

$$\lesssim |(\gamma - \tilde{\gamma})_{s,t}| + (|\tilde{\gamma}_{s,t}| + |\gamma_{s,t}|) \|\gamma - \tilde{\gamma}\|_\alpha$$

$$\text{So } \|f(\gamma) - f(\tilde{\gamma})\|_\alpha \lesssim \|\gamma - \tilde{\gamma}\|_\alpha + (\|\gamma\|_\alpha + \|\tilde{\gamma}\|_\alpha) \|\gamma - \tilde{\gamma}\|_\alpha$$

$$\|\gamma - \tilde{\gamma}\|_\alpha \leq |\gamma_0 - \tilde{\gamma}_0| + \|\gamma - \tilde{\gamma}\|_\alpha T^\alpha \lesssim \|\gamma - \tilde{\gamma}\|_\alpha + (\|\gamma\|_\alpha + \|\tilde{\gamma}\|_\alpha) (|\gamma_0 - \tilde{\gamma}_0| + \|\gamma - \tilde{\gamma}\|_\alpha)$$

$$= \|\gamma - \tilde{\gamma}\|_\alpha (1 + \|\gamma\|_\alpha + \|\tilde{\gamma}\|_\alpha) + |\gamma_0 - \tilde{\gamma}_0| (\|\gamma\|_\alpha + \|\tilde{\gamma}\|_\alpha)$$

$$\leq (1 + \|\gamma\|_\alpha + \|\tilde{\gamma}\|_\alpha) (\|\gamma - \tilde{\gamma}\|_\alpha + |\gamma_0 - \tilde{\gamma}_0|)$$

□

Existence and uniqueness of the Yang DE: let $\beta \in (\frac{1}{2}, 1]$ and let $X \in C^\beta([0, T]; \mathbb{R}^d)$.

Let $f \in C_b^2(\mathbb{R}^m; 2(\mathbb{R}^d; \mathbb{R}^m))$ and $y \in \mathbb{R}^m$. Then, $\exists! Y \in C^\beta([0, T]; \mathbb{R}^m)$ which satisfies

$\hat{f}$
boundedness
is necessary
(see Hairer-Fritz
p. 133)

$$Y_t = y + \int_0^t f(Y_s) dX_s \quad \forall t \in [0, T].$$

$\uparrow$
for the previous lemma

$f(Y) \in C^\beta \Rightarrow$ the Yang integral is well-defined

Proof: we want to use the contraction theorem.

Let $\alpha \in (\frac{1}{2}, \beta)$. We define the map $\Pi_t: C^\alpha([0, t]; \mathbb{R}^m) \rightarrow C^\alpha([0, T]; \mathbb{R}^m)$

$$\Pi_t(Y) = y + \int_0^t f(Y_s) dX_s$$

Let $B_t := \{Y \in C^\alpha([0, t]; \mathbb{R}^m) : Y_0 = y, \|Y\|_{\alpha; [0, t]} \leq 1\}$ which is a complete metric space since it is a closed subset of the Banach space $C^\alpha([0, t]; \mathbb{R}^m)$ with seminorm $\|\cdot\|_{\alpha; [0, t]}$.

Invariance: we want to see that if $Y \in B_t \Rightarrow \Pi_t(Y) \in B_t$ (choosing a proper t)

$$\|\Pi_t(Y)\|_{\alpha; [0, t]} = \left\| \int_0^t f(Y_s) dX_s \right\|_{\alpha; [0, t]} \leq C(|y| + \|Y\|_{\alpha; [0, t]}) \|X\|_{\alpha; [0, t]}$$

$\uparrow$
use the estimate in the
continuity of Yang integral
with $\tilde{Y} = \tilde{X} = 0$

$$\begin{aligned} \left\| \int_0^t f(Y_s) dX_s \right\|_{\alpha; [0, t]} &\leq C(|y| + \|Y\|_{\alpha; [0, t]}) \|X\|_{\alpha; [0, t]} \\ &\leq C \|X\|_{\alpha; [0, t]} \leq 2 \|X\|_{\beta; [0, t]} t^{\beta-\alpha} \leq 2 \|X\|_{\beta; [0, t]} t^{\beta-\alpha} \end{aligned}$$

$\uparrow$
this constant $\|Y\|_{\alpha; [0, t]} \leq 1$

$\hat{C}$
it will be important
for giving the
solution

Choosing $t = t_1 > 0$ sufficiently small s.t. $2C \|X\|_{\beta; [0, t]} t^{\beta-\alpha} \leq 1$ we have

$\|\Pi_{t_1}(Y)\|_{\alpha; [0, t_1]} \leq 1$. Furthermore $\Pi_{t_1}(Y)_0 = y \Rightarrow \Pi_{t_1}: B_{t_1} \rightarrow B_{t_1}$ is well-defined.

Contraction: let $Y, \tilde{Y} \in B_t$: (they start at the same point so the norm is actually $\|\cdot\|_{\alpha; [0, t]}$):

$$\|\Pi_t(Y) - \Pi_t(\tilde{Y})\|_{\alpha; [0, t]} = \left\| \int_0^t f(Y_s) dX_s - \int_0^t f(\tilde{Y}_s) dX_s \right\|_{\alpha; [0, t]}$$

estimate in the
continuity of
Yang integral

$$\rightarrow \leq \|f(Y) - f(\tilde{Y})\|_{\alpha; [0, t]} \|X\|_{\alpha; [0, t]}$$

previous Lemma

$$\rightarrow \leq \|Y - \tilde{Y}\|_{\alpha; [0, t]} \|X\|_{\alpha; [0, t]}$$

$$+ \|Y\|_{\alpha; [0, t]} \leq 1$$

$$\|\tilde{Y}\|_{\alpha; [0, t]} \leq 1$$

$$\leq \|Y - \tilde{Y}\|_{\alpha; [0, t]} \|X\|_{\beta; [0, t]} t^{\beta-\alpha} \leq \|Y - \tilde{Y}\|_{\alpha; [0, t]} \|X\|_{\beta; [0, t]} t^{\beta-\alpha}$$

So taking $t = t_2 \in (0, t_1)$ s.t. $\frac{1}{C} \|X\|_{\beta; [0, t]} t^{\beta-\alpha} \leq \frac{1}{2} \leftarrow$ the constant given by $\hat{C}$ we need the strict " $<$ "

we have $\|M_t(Y) - M_t(\tilde{Y})\|_{\alpha; [0, t_2]} \leq \frac{1}{2} \|Y - \tilde{Y}\|_{\alpha; [0, t_2]} < \|Y - \tilde{Y}\|_{\alpha; [0, t_2]}$

we need the strict

So, for the Banach fixed point theorem $\exists! Y$

fixed point of M_{t_2} on B_{t_2} .

Glue the solution together: consider the Young DE:

if $t_2 \leq T$ no problem.
otherwise we take a min

$$\tilde{Y}_u = Y_{t_2} + \int_0^u f(\tilde{Y}_s) d\tilde{X}_s \quad u \in [0, t_2] \quad \text{with} \quad \tilde{X}_s := X_{t_2+s} - X_{t_2}$$

We can do the same as before, since C and $\bar{C}$ does not depend on the initial data,

$$\|\tilde{X}\|_{\beta, [0, t_2]} = \|X\|_{\beta, [t_2, 2t_2]} \leq \|X\|_{\beta} \quad \text{so the estimates don't change and}$$

we find the unique $\tilde{Y}_u$ defined on $[0, t_2]$. Now I define $Y_t := \tilde{Y}_{t-t_2} \quad \forall t \in [t_2, 2t_2]$,

we want to see that Y_t is a solution of the starting DE between $[t_2, 2t_2]$:

$$\forall t \in [t_2, 2t_2]: Y_t = \tilde{Y}_{t-t_2} = Y_{t_2} + \int_0^{t-t_2} f(\tilde{Y}_s) d\tilde{X}_s =$$

$$d\tilde{X}_s = dX_{s+t_2}$$

$$Y_{t_2} \text{ solve the DE} \rightarrow = \left(y + \int_0^{t_2} f(Y_s) dX_s \right) + \int_0^{t-t_2} f(Y_{s+t_2}) dX_{s+t_2}$$

$$\text{2nd integral: } u := s+t_2 \rightarrow = y + \int_0^{t_2} f(Y_s) dX_s + \int_{t_2}^t f(Y_u) dX_u = y + \int_0^t f(Y_s) dX_s$$

$s=0 \Rightarrow u=t_2$
 $s=t-t_2 \Rightarrow u=t$

The solution is C^β : for now, our solution Y is in C^α . However $X \in C^\beta$ so

$$|Y_{s,t}| = \left| \int_s^t f(Y_s) dX_s \right| \leq \|f\|_{\infty} \|X\|_{\beta} |t-s|^\beta + C \|Y\|_{\alpha} \|X\|_{\beta} |t-s|^{\alpha+\beta}$$

$\uparrow$
 $Y \text{ sol.}$

$\uparrow$
estimate in the existence of Young Int.

$$\Rightarrow \|Y\|_{\beta} \leq \|Y\|_{\infty} \|X\|_{\beta} + C \|Y\|_{\alpha} \|X\|_{\beta} T < \infty.$$

So the solution is in C^β , obviously if another solution exists in $C^\beta \Rightarrow$ is in C^α and by

the uniqueness in C^α we have finished. $\square$

Continuity w.r.t. the data: let $p \in (\frac{1}{2}, 1]$ and $f \in C_b^1$. Let $X, \tilde{X} \in C^\beta$ and $y, \tilde{y} \in \mathbb{R}^m$.

Let $Y, \tilde{Y}$ the unique solutions of the Young DE's with data (y, X) and $(\tilde{y}, \tilde{X})$.

Let $M > 0$ be a constant s.t. $\|X\|_\beta, \|\tilde{X}\|_\beta \leq M$. Then, $\forall \alpha \in (\frac{1}{2}, \beta)$, there exists a constant

$C_M > 0$ depending on $\alpha, T, \|f\|_{C_b^1}$ and M s.t.

$$\|Y - \tilde{Y}\|_\alpha \leq C_M (|y - \tilde{y}| + \|X - \tilde{X}\|_\alpha)$$

Proof: from the proof of the previous theorem the local sol. Y over $[0, t_2]$ is an element

of B_{t_2} , in particular $\|Y\|_{\alpha; [0, t_2]} \leq 1$, similarly $\|\tilde{Y}\|_{\alpha; [0, \tilde{t}_2]} \leq 1$ for $\tilde{t}_2 > 0$.

Then $\forall t \in [0, t_2 \wedge \tilde{t}_2]$ we have:

$$\|Y - \tilde{Y}\|_{\alpha; [0, t]} = \left\| \int_0^\cdot f(Y_s) dX_s - \int_0^\cdot f(\tilde{Y}_s) d\tilde{X}_s \right\|_{\alpha; [0, t]}$$

estimates from the cont. of Young int. $\rightarrow \lesssim (|f(Y_0) - f(\tilde{Y}_0)| + \|f(Y) - f(\tilde{Y})\|_{\alpha; [0, t]}) \|X\|_{\alpha; [0, t]} + (|f(\tilde{Y})| + \|f(\tilde{Y})\|_{\alpha; [0, t]}) \|X - \tilde{X}\|_{\alpha; [0, t]}$

lemma at the beginning $\rightarrow \lesssim (|Y_0 - \tilde{Y}_0| + \|Y - \tilde{Y}\|_{\alpha; [0, t]}) \|X\|_{\alpha; [0, t]} + \|X - \tilde{X}\|_{\alpha; [0, t]}$
 $+ \|Y\|_{\alpha; [0, t]} \leq 1$
 $\|Y\|_{\alpha; [0, t]} \leq 1$

$$\leq (|Y_0 - \tilde{Y}_0| + \|Y - \tilde{Y}\|_{\alpha; [0, t]}) \|X\|_{\beta; [0, t]} t^{\beta-\alpha} + \|X - \tilde{X}\|_{\alpha; [0, T]}$$

Choosing $t = t_3 \in (0, t_2 \wedge \tilde{t}_2]$ sufficiently small s.t. $\underbrace{C_2 \|X\|_{\beta; [0, t]}}_{\text{the constant from } \lesssim} t^{\beta-\alpha} \leq \frac{1}{2}$ we have

$$\|Y - \tilde{Y}\|_{\alpha; [0, t_3]} \lesssim |Y_0 - \tilde{Y}_0| + \|X - \tilde{X}\|_{\alpha; [0, t_3]} \quad (*)$$

Now, doing the same trick used to glue together the solution of Young DE's, we can define translations

of $Y, X, \tilde{Y}, \tilde{X}$ s.t. the translations solve the same Young DE's but with starting

here somewhere we use the bound M

point Y_s and $\tilde{Y}_s$ so we have the same estimate (*) for those translation $(\hat{Y}, \hat{\tilde{Y}}, \hat{X}, \hat{\tilde{X}})$.

$$\text{But } \|\hat{Y} - \hat{\tilde{Y}}\|_{\alpha; [0, t-s]} = \|Y - \tilde{Y}\|_{\alpha; [s, t]} \quad \text{and} \quad \|\hat{X} - \hat{\tilde{X}}\|_{\alpha; [0, t-s]} = \|X - \tilde{X}\|_{\alpha; [s, t]}$$

and so $\forall s, t \in [0, T]$ s.t. $|t-s| < t_3$ (when we choose $t = t_3$ we could choose every $t < t_3$)

$$\|Y - \tilde{Y}\|_{\alpha; [s, t]} \lesssim |Y_s - \tilde{Y}_s| + \|X - \tilde{X}\|_{\alpha; [s, t]} \quad (*)$$

Now we take a partition π of $[0, t]$ s.t. $|\pi| \leq t_3$. Taking a node t_{i+1} we have

$$|Y_{t_{i+1}} - \tilde{Y}_{t_{i+1}}| \leq |Y_{t_i} - \tilde{Y}_{t_i}| + \|Y - \tilde{Y}\|_{\alpha; [t_i, t_{i+1}]} (t_{i+1} - t_i)^\alpha$$

$\uparrow$
def of $\|\cdot\|_\alpha$

$$\begin{aligned}
 (\star) \quad & \lesssim |Y_{t_i} - \tilde{Y}_{t_i}| + \left((|Y_{t_i} - \tilde{Y}_{t_i}| + \|X - \tilde{X}\|_{\alpha; [t_i, t_{i+1}]}) (t_{i+1} - t_i)^\alpha \right) \\
 & = \underbrace{|Y_{t_i} - \tilde{Y}_{t_i}|}_{\substack{\text{Iterate on this} \\ \text{and to obtain the result}}} \left(1 + (t_{i+1} - t_i)^\alpha \right) + \|X - \tilde{X}\|_{\alpha; [t_i, t_{i+1}]} (t_{i+1} - t_i)^\alpha
 \end{aligned}$$

□

Functions of controlled paths

Lemma: $\alpha \in (\frac{1}{3}, \frac{1}{2}]$ and $X \in C^\alpha$. Let $f \in C_b^2$. For any $(Y, Y') \in \mathcal{D}_X^{2\alpha}$ the pair

$$(f(Y), Df(Y)Y') \in \mathcal{D}_X^{2\alpha}$$

Moreover we have the estimates

$$\|Df(Y)Y'\|_\alpha \leq C (1 + |Y'_0| + \|Y'\|_\alpha + \|R^Y\|_{2\alpha})^2 (1 + \|X\|_\alpha)$$

$$\|R^{f(Y)}\|_{2\alpha} \leq C (1 + |Y'_0| + \|Y'\|_\alpha + \|R^Y\|_{2\alpha})^2 (1 + \|X\|_\alpha)^2$$

where the constant C depends on α, T and $\|f\|_{C_b^2}$.

Proof: 1° inequality:

$$\begin{aligned}
 |Df(Y_t)Y'_t - Df(Y_s)Y'_s| & \leq \underbrace{|Df(Y_t)Y'_t - Df(Y_t)Y'_s|}_{\leq \|Df(Y_t)\|_\alpha \|Y'_t - Y'_s\|_\alpha} + |Df(Y_t) - Df(Y_s)| |Y'_s| \\
 & \lesssim |Y'_{s,t}| + \left| \int_0^1 D^2f(Y_s + \theta(Y_t - Y_s))(Y_t - Y_s) d\theta \right| |Y'_s| \\
 & \lesssim |Y'_{s,t}| + |Y_{s,t}| |Y'_s|
 \end{aligned}$$

So:

$$\|Df(Y)Y'\|_\alpha \lesssim \|Y'\|_\alpha + \|Y\|_\alpha \|Y'\|_\alpha$$

$$\frac{Y_{s,t}}{|t-s|^\alpha} = \frac{Y'_s}{|t-s|^\alpha} \frac{X_{s,t}}{|t-s|^\alpha} + \frac{R^Y_{s,t}}{|t-s|^{2\alpha}} \cdot |t-s|^\alpha \lesssim \|Y'\|_\alpha + \left(\|Y'\|_\infty \|X\|_\alpha + \|R^Y\|_{2\alpha} \right) \|Y'\|_\infty$$

$$\|Y'\|_\infty \leq |Y'_0| + \|Y'\|_\alpha T \longrightarrow \lesssim \|Y'\|_\alpha + \left((|Y'_0| + \|Y'\|_\alpha) \|X\|_\alpha + \|R^Y\|_{2\alpha} \right) (|Y'_0| + \|Y'\|_\alpha)$$

it's very simple $\longrightarrow = \|Y'\|_\alpha + (|Y'_0| + \|Y'\|_\alpha)^2 \|X\|_\alpha + \|R^Y\|_{2\alpha} (|Y'_0| + \|Y'\|_\alpha)$

to see the the

3 terms of the the upper line are also in the second line $\lesssim (1 + |Y'_0| + \|Y'\|_\alpha + \|R^Y\|_{2\alpha})^2 (1 + \|X\|_\alpha)$

2° inequality : $R_{s,t}^{f(Y)} := f(Y_t) - f(Y_s) - Df(Y_s) Y'_s X_{s,t}$

def. $R_{s,t}^Y \rightarrow = f(Y_t) - f(Y_s) - Df(Y_s)(X_{s,t} - R_{s,t}^Y)$

Taylor expansion $\rightarrow = \int_0^1 \int_0^1 D^2 f(Y_s + r_1 r_2 Y_{s,t}) Y_{s,t}^{\otimes 2} r_1 dr_2 dr_1 + Df(Y'_s) R_{s,t}^Y$

So $|R_{s,t}^{f(Y)}| \lesssim |Y_{s,t}|^2 + |R_{s,t}^Y| \Rightarrow \|R_{s,t}^{f(Y)}\|_{2\alpha} \lesssim \|Y\|_\alpha^2 + \|R^Y\|_{2\alpha}$

$Y_{s,t} = Y'_s X_{s,t} + R_{s,t}^Y \rightarrow \lesssim (\|Y'\|_\infty \|X\|_\alpha + \|R^Y\|_{2\alpha})^2 + \|R^Y\|_{2\alpha}$
 $\lesssim (1 + |Y'_0| + \|Y'\|_\alpha + \|R^Y\|_{2\alpha})^2 (1 + \|X\|_\alpha)^2$

□

Lemma : let $\alpha \in (\frac{1}{3}, \frac{1}{2}]$; $X, \tilde{X} \in C^\alpha$; $Y, Y' \in D_X^{2\alpha}$, $(\tilde{Y}, \tilde{Y}') \in D_X^{2\alpha}$ and $f \in C_b^3$.

Let $M > 0$ s.t. $\|X\|_\alpha \leq M$, $\|\tilde{X}\|_\alpha \leq M$, $|Y'_0| + \|Y'\|_\alpha + \|R^Y\|_{2\alpha} \leq M$ and

$|\tilde{Y}'_0| + \|\tilde{Y}'\|_\alpha + \|R^{\tilde{Y}}\|_{2\alpha} \leq M$. Then, $\exists C$ constant depending $\alpha, T, \|f\|_{C_b^3}$ and M s.t. :

$\cdot \|Df(Y)Y' - Df(\tilde{Y})\tilde{Y}'\| \leq C (|Y'_0 - \tilde{Y}'_0| + |Y'_0 - \tilde{Y}'_0| + \|Y' - \tilde{Y}'\|_\alpha + \|R^Y + R^{\tilde{Y}}\|_{2\alpha} + \|X - \tilde{X}\|_\alpha)$

$\cdot \|R^{f(Y)} - R^{f(\tilde{Y})}\|_{2\alpha} \leq C (|Y'_0 - \tilde{Y}'_0| + |Y'_0 - \tilde{Y}'_0| + \|Y - \tilde{Y}\|_\alpha + \|R^Y - R^{\tilde{Y}}\|_{2\alpha} + \|X - \tilde{X}\|_\alpha)$

Proof Now we use $\lesssim$ also to indicate the dependence of C on M .

First estimate :

$| (Df(Y)Y' - Df(\tilde{Y})\tilde{Y}')_{s,t} | \leq \left| Df(Y)Y' \right|_{s,t} \pm \left| Df(Y)\tilde{Y}' \right|_{s,t}$
 $\leq | (Df(Y)(Y' - \tilde{Y}'))_{s,t} | + | (Df(\tilde{Y}) - Df(Y))\tilde{Y}' |_{s,t} |$
 $\pm Df(Y_t)(Y'_s - \tilde{Y}'_s)$
 in the 1st l.i., $\rightarrow = | Df(Y_t)(Y' - \tilde{Y}')_{s,t} | + | Df(Y)_{s,t}(Y'_s - \tilde{Y}'_s) | +$
 $\pm (Df(Y_t) - Df(\tilde{Y}_t))\tilde{Y}'_s$
 in the 2nd l.i. $+ | (Df(Y_t) - Df(\tilde{Y}_t))\tilde{Y}'_{s,t} | + | (Df(Y) - Df(\tilde{Y}))_{s,t} Y'_s |$

Hence

$\|Df(Y)Y' - Df(\tilde{Y})\tilde{Y}'\|_\alpha \leq \|Df(Y_t)\|_\infty \|Y' + \tilde{Y}'\|_\alpha + \|Df(Y)\|_\alpha \|Y' - \tilde{Y}'\|_\infty$
 $+ \|Df(Y) - Df(\tilde{Y})\|_\infty \|\tilde{Y}'\|_\alpha + \|Df(Y) - Df(\tilde{Y})\|_\alpha \|Y'\|_\infty$

$\|Df(Y)\|_\alpha \leq \frac{1}{2} \|D^2 f\|_\infty \|Y\|_\alpha \xrightarrow{\text{see the proof}} \lesssim \|Y' + \tilde{Y}'\|_\alpha + \|Y' - \tilde{Y}'\|_\infty + \|Y - \tilde{Y}\|_\infty + \|Df(Y) - Df(\tilde{Y})\|_\alpha$
 that $(f(Y), Df(Y))$ is controlled and the same $\|Df(Y) - Df(\tilde{Y})\|_\infty \leq \|D^2 f\|_\infty \|Y - \tilde{Y}\|_\infty$

$Df \in C_b^2$ so I can use the $\rightarrow \lesssim \|Y' - \tilde{Y}'\|_\alpha + \|Y' - \tilde{Y}'\|_{\mathbb{W}} + \|Y - \tilde{Y}\|_\infty + |Y_0 - \tilde{Y}_0| + \|Y - \tilde{Y}\|_\alpha$
 1st lemma in the Young DE
 section for $\|Df(Y) - Df(\tilde{Y})\|_\alpha$

result on the estimate $\|Y - \tilde{Y}\|_\alpha \lesssim \rightarrow \lesssim$ result.

$$\lesssim |Y_0 - \tilde{Y}_0| + \|Y' - \tilde{Y}'\|_\alpha + \|X - \tilde{X}\|_\alpha + \|R^Y - R^{\tilde{Y}}\|_{2\alpha}$$

Second estimate:

$$\begin{aligned} |R_{s,t}^{f(Y)} - R_{s,t}^{f(\tilde{Y})}| &= |f(Y)_{s,t} - Df(Y_s) Y_s' X_{s,t} - f(\tilde{Y})_{s,t} + Df(\tilde{Y}_s) \tilde{Y}_s' X_{s,t}| \\ &= |f(Y)_{s,t} - Df(Y_s)(Y_{s,t} - R_{s,t}^Y) - f(\tilde{Y})_{s,t} + Df(\tilde{Y}_s)(\tilde{Y}_{s,t} - R_{s,t}^{\tilde{Y}})| \\ &\leq |f(Y)_{s,t} - Df(Y_s) Y_{s,t} - f(\tilde{Y})_{s,t} + Df(\tilde{Y}_s) \tilde{Y}_{s,t}| + |Df(Y_s) R_{s,t}^Y - Df(\tilde{Y}_s) R_{s,t}^{\tilde{Y}}| \\ &= \left| \int_0^1 \int_0^1 D^2 f(Y_s + r_1 r_2 Y_{s,t}) Y_{s,t}^{\otimes 2} r_1 dr_2 dr_1 - \int_0^1 \int_0^1 D^2 f(\tilde{Y}_s + r_1 r_2 \tilde{Y}_{s,t}) \tilde{Y}_{s,t}^{\otimes 2} r_1 dr_2 dr_1 \right| \\ &\quad + |Df(Y_s) R_{s,t}^Y - Df(\tilde{Y}_s) R_{s,t}^{\tilde{Y}}| \end{aligned}$$

put together the two $\int_0^1 \int_0^1$ and

$$\begin{aligned} \pm D^2 f(Y_s + r_1 r_2 Y_{s,t}) Y_{s,t}^{\otimes 2} &\leq \int_0^1 \int_0^1 |D^2 f(Y_s + r_1 r_2 Y_{s,t}) - D^2 f(\tilde{Y}_s + r_1 r_2 \tilde{Y}_{s,t})| |Y_{s,t}^{\otimes 2}| + \|D^2 f\|_\infty |Y_{s,t}^{\otimes 2} - \tilde{Y}_{s,t}^{\otimes 2}| r_1 dr_2 dr_1 \\ &\quad + |Df(Y_s) R_{s,t}^Y - Df(\tilde{Y}_s) R_{s,t}^{\tilde{Y}}| \end{aligned}$$

(*) we use

$$\begin{aligned} |D^2 f(a) - D^2 f(b)| &\lesssim \|Y - \tilde{Y}\|_\infty |Y_{s,t}|^2 + |Y_{s,t}^{\otimes 2} - \tilde{Y}_{s,t}^{\otimes 2}| + |Df(Y_s) R_{s,t}^Y - Df(\tilde{Y}_s) R_{s,t}^{\tilde{Y}}| \\ &\lesssim \|Y - \tilde{Y}\|_\infty |Y_{s,t}|^2 + |Y_{s,t}| |Y_{s,t} - \tilde{Y}_{s,t}| + |\tilde{Y}_{s,t}| |Y_{s,t} - \tilde{Y}_{s,t}| \\ &\quad + |Df(Y_s) - Df(\tilde{Y}_s)| |R_{s,t}^Y| + |Df(\tilde{Y}_s)| |R_{s,t}^Y - R_{s,t}^{\tilde{Y}}| \end{aligned}$$

so that

$$\begin{aligned} \|R^{f(Y)} - R^{f(\tilde{Y})}\|_{2\alpha} &\lesssim \|Y - \tilde{Y}\|_\infty \|Y\|_\alpha^2 + (\|Y\|_\alpha + \|\tilde{Y}\|_\alpha) \|Y - \tilde{Y}\|_\alpha + \|Y - \tilde{Y}\|_\infty \|R^Y\|_{2\alpha} + \|R^Y - R^{\tilde{Y}}\|_{2\alpha} \\ &\lesssim |Y_0 - \tilde{Y}_0| + \|Y - \tilde{Y}\|_\alpha + \|R^Y - R^{\tilde{Y}}\|_{2\alpha} \end{aligned}$$

$\lesssim$ result.

□

Lemma: let $\alpha \in (\frac{1}{3}, \frac{1}{2}]$ and $\mathbf{X} = (X, X) \in \mathcal{C}^\alpha$. Let $f \in C_b^2$. For any $(Y, Y') \in \mathcal{D}_X^{2\alpha}$, the pair

$$\left(\int_0^\cdot f(Y_u) d\mathbf{X}_u, f(Y) \right) \in \mathcal{D}_X^{2\alpha}$$

is a controlled path.

Moreover we have the estimates:

$$\|f(Y)\|_\alpha \leq C \left(\|Y_0'\| + \|Y'\|_\alpha \right) \|X\|_\alpha + \|R^Y\|_{2\alpha} T^\alpha$$

$$\|R \int_0^\cdot f(Y_u) d\mathbf{X}_u\|_{2\alpha} \leq C \left(1 + \|Y_0'\| + \|Y'\|_\alpha + \|R^Y\|_{2\alpha} \right)^2 \left(1 + \|X\|_\alpha \right)^2 \|\mathbf{X}\|_\alpha$$

where the constant C depends on α, T and $\|f\|_{C_b^2}$.

Proof: Well defined integral: for a previous lemma $(f(Y), Df(Y)Y')$ is a controlled path

and so $\int_0^\cdot f(Y_u) d\mathbf{X}_u$ is well defined.

1° estimate:

$$|f(Y)_{s,t}| \leq \underbrace{\|Df\|_\infty}_{\text{Lipsch.}} |Y'_{s,t}| = \|Df\|_\infty |Y'_s X_{s,t} + R^Y_{s,t}|$$

$$\text{So } \|f(Y)\|_\alpha \lesssim \|Y'\|_\infty \|X\|_\alpha + \|R^Y\|_{2\alpha} T^\alpha$$

$$\leq (\|Y_0'\| + \|Y'\|_\alpha T^\alpha) \|X\|_\alpha + \|R^Y\|_{2\alpha} T^\alpha$$

$$\leq (1 + T^\alpha) (\|Y_0'\| + \|Y'\|_\alpha) \|X\|_\alpha + \|R^Y\|_{2\alpha} T^\alpha$$

2° estimate: for an estimate in the rough integral section:

$$\left| R \int_{s,t} f(Y_u) d\mathbf{X}_u \right| \stackrel{\text{def}}{=} \left| \int_s^t f(Y_u) d\mathbf{X}_u - f(Y_s) X_{s,t} \right|$$

$$\leq \left| \int_s^t f(Y_u) d\mathbf{X}_u - f(Y_s) X_{s,t} - Df(Y_s) Y'_s X_{s,t} \right| + |Df(Y_s) Y'_s X_{s,t}|$$

$$\lesssim \left(\|R^{f(Y)}\|_{2\alpha} \|X\|_\alpha + \|Df(Y)Y'\|_\alpha \|X\|_{2\alpha} \right) |t-s|^{3\alpha} + |Df(Y_s) Y'_s X_{s,t}|$$

So

$$\|R \int_{s,t} f(Y_u) d\mathbf{X}_u\|_{2\alpha} \lesssim \underbrace{\|R^{f(Y)}\|_{2\alpha}}_{\text{we use the estimates in the 1st lemma of this section}} \|X\|_\alpha + \|Df(Y)Y'\|_\alpha \|X\|_{2\alpha} + \|Y'\|_\infty \|X\|_{2\alpha}$$

$\lesssim$ result. $\square$

Lemma: let $\alpha \in (\frac{1}{3}, \frac{1}{2}]$ and $\mathbf{X} = (X, \mathbb{X})$, $\tilde{\mathbf{X}} = (\tilde{X}, \tilde{\mathbb{X}}) \in \mathcal{C}^\alpha$. Let $(Y, Y') \in \mathcal{D}_X^{2\alpha}$

and $(\tilde{Y}, \tilde{Y}') \in \mathcal{D}_X^{2\alpha}$ and $f \in C_b^3$. Let $M > 0$ be a constant s.t. $\|X\|_\alpha, \|\tilde{X}\|_\alpha \leq M$,

$|Y_0| + \|Y'\|_\alpha + \|R^Y\|_{2\alpha} \leq M$, $|\tilde{Y}_0| + \|\tilde{Y}'\|_\alpha + \|R^{\tilde{Y}}\|_{2\alpha} \leq M$. Then

$$\|f(Y) - f(\tilde{Y})\|_\alpha \leq C(|Y_0 - \tilde{Y}_0| + (|Y_0' - \tilde{Y}_0'| + \|Y' - \tilde{Y}'\|_\alpha) \|X\|_\alpha + \|R^Y - R^{\tilde{Y}}\|_{2\alpha} T^\alpha + \|X - \tilde{X}\|_\alpha)$$

$$\|R \int_0^\cdot f(Y_s) d\mathbf{X}_s - R \int_0^\cdot f(\tilde{Y}_s) d\tilde{\mathbf{X}}_s\|_{2\alpha} \leq C(|Y_0 - \tilde{Y}_0| + |Y_0' - \tilde{Y}_0'| + \|\tilde{Y} - Y\|_\alpha + \|R^Y - R^{\tilde{Y}}\|_{2\alpha} + \|X - \tilde{X}\|_\alpha) (\|\mathbf{X}\|_\alpha + \|\tilde{\mathbf{X}}\|_\alpha)$$

→ Proof: we combine the estimates in the section of rough integration for $Y - \tilde{Y}$ and $R^Y - R^{\tilde{Y}}$

with the lemmas of this section. $\square$

Rough differential equation

Existence and uniqueness of the RDE: let $\beta \in (\frac{1}{3}, \frac{1}{2}]$ and let $\mathbf{X} = (X, \mathbb{X}) \in \mathcal{C}^\beta([0, T]; \mathbb{R}^d)$

a rough path. Let $f \in C_b^3(\mathbb{R}^m; \mathcal{L}(\mathbb{R}^d; \mathbb{R}^m))$ and let $y \in \mathbb{R}^m$. Then, $\exists!$ controlled path

$(Y, Y') \in \mathcal{D}_X^{2\beta}([0, t]; \mathbb{R}^m)$ s.t. $Y' = f(Y)$ and

↑ it makes sense this request since we know $(\int_0^\cdot f(Y) d\mathbf{X}_s, f(Y))$ is a controlled path

$$Y_t = y + \int_0^t f(Y_s) d\mathbf{X}_s \quad \forall t \in [0, T]$$

→ Proof: let $\alpha \in (\frac{1}{3}, \beta)$. Define the map $\Pi_t: \mathcal{D}_X^{2\alpha}([0, t]; \mathbb{R}^m) \rightarrow \mathcal{D}_X^{2\alpha}([0, t]; \mathbb{R}^m)$ by

$$\Pi_t(Y, Y') := \left(y + \int_0^\cdot f(Y_s) d\mathbf{X}_s, f(Y) \right)$$

which defines a control path for the previous section. Let

$$B_t := \{(Y, Y') \in \mathcal{D}_X^{2\alpha}([0, t]; \mathbb{R}^m) : Y_0 = y, Y_0' = f(y), \|Y, Y'\|_{X, \alpha, [0, t]} \leq 1\}$$

where $\|\cdot\|_{X, \alpha}$ it's the usual seminorm of $\mathcal{D}_X^{2\alpha}$: $\|Y, Y'\|_{X, \alpha, [0, t]} = \|Y'\|_\alpha + \|R^Y\|_{2\alpha}$

In particular in B_t the starting points are fixed so it is a real norm and since is a closed

subspace of the Banach space $\mathcal{D}_X^{2\alpha}$ with norm $|Y_0| + |Y_0'| + \|Y'\|_\alpha + \|R^Y\|_{2\alpha}$ (that is equivalent

to $\|\cdot\|_{X, \alpha}$ on B_t for what we have said) it is itself a Banach space.

B_t is non empty: $(Y_s := y + f(y) X_{0,s}, Y_s' := f(y)) \in B_t \quad \forall t$
↑
is constant!

Choose $\mathcal{M}$ to use 1st lemma of previous section: $\mathcal{M} := 1 + \|f\|_{C_b^3} + \|X\|_{\alpha; [a, T]}$.

Indeed if $(Y, Y') \in B_t$ then $|Y_0'| + \|Y'\|_{\alpha} + \|R^Y\|_{2\alpha} = |f(y)| + \|Y, Y'\|_{X, \alpha}$

$$\leq \|f\|_{C_b^3} + 1 \leq \mathcal{M},$$

In the following δ depends also on $\mathcal{M}$.

Invariance: let $(Y, Y') \in B_t$.

$$\|M_t(Y, Y')\|_{X, \alpha, [0, t]} \stackrel{\text{def}}{=} \|f(Y)\|_{\alpha; [0, t]} + \|R^{\int_0^\cdot f(Y_s) dX_s}\|_{2\alpha; [0, t]}$$

estimate previous lemma $\rightarrow \lesssim (|Y_0'| + \|Y'\|_{\alpha; [0, t]}) \|X\|_{\alpha; [0, t]} + \|R^Y\|_{2\alpha; [0, t]} t^\alpha +$

$$+ (1 + |Y_0'| + \|Y'\|_{\alpha; [0, t]} + \|R^Y\|_{2\alpha; [0, t]})^2 (1 + \|X\|_{\alpha; [0, t]})^2 \|X\|_{\alpha; [0, t]}$$

previous hp. on $\mathcal{M} \rightarrow \lesssim \|X\|_{\alpha; [0, t]} + t^\alpha$

$$= \|X\|_{\alpha; [0, t]} + \|X\|_{2\alpha; [0, t]} + t^\alpha$$

$$\leq \|X\|_{\beta; [0, t]} t^{\beta-\alpha} + \|X\|_{2\beta; [0, t]} t^{2(\beta-\alpha)} + t^\alpha$$

So we can choose $t = t_1$ small enough s.t. $\|M_{t_1}(Y, Y')\|_{X, \alpha, [0, t_1]} \leq 1$ and so

B_t is invariant.

Contraction: let $(Y, Y'), (\tilde{Y}, \tilde{Y}') \in B_t$ for some $t \in (0, t_1]$. So

$$\|M_t(Y, Y') - M_t(\tilde{Y}, \tilde{Y}')\|_{X, \alpha, [0, t]} = \|f(Y) - f(\tilde{Y})\|_{\alpha; [0, t]} + \|R^{\int_0^\cdot f(Y_s) dX_s} - R^{\int_0^\cdot f(\tilde{Y}_s) dX_s}\|_{2\alpha; [0, t]}$$

last lemma previous section $\rightarrow \lesssim (\|Y' - \tilde{Y}'\|_{\alpha; [0, t]} + \|R^Y - R^{\tilde{Y}}\|_{2\alpha; [0, t]}) (\|X\|_{\alpha; [0, t]} + t^\alpha)$

($\tilde{X} = X$ and $(Y, Y') - (\tilde{Y}, \tilde{Y}')$ starts at the same points)

$$\leq (\|Y' - \tilde{Y}'\|_{\alpha; [0, t]} + \|R^Y - R^{\tilde{Y}}\|_{2\alpha; [0, t]}) (\|X\|_{\beta; [0, t]} t^{\beta-\alpha} + \|X\|_{2\beta; [0, t]} t^{2(\beta-\alpha)} + t^\alpha)$$

$$=: \|(Y, Y') - (\tilde{Y}, \tilde{Y}')\|_{X, \alpha, [0, t]}$$

So taking $t := t_2 \in (0, t_1]$ sufficiently small we have

$$\|M_t(Y, Y') - M_t(\tilde{Y}, \tilde{Y}')\|_{X, \alpha, [0, t]} \leq \frac{1}{2} \|(Y, Y') - (\tilde{Y}, \tilde{Y}')\|_{X, \alpha, [0, t]}$$

So M_t is a contraction and we have a unique fixed point $(Y, Y') \in \mathcal{D}_X^{\alpha}$ of the map

M_t and so it's the only controlled in $\mathcal{D}_X^{\alpha}$ that satisfy the RDE on $[0, t_2]$ with $Y' = f(\cdot)$.

Glue the solution together: we use the same translation on time used in the uniqueness of "rough

exponential" to define a new RDE starting in Y_{t_2} . Since all the constants do not depend

on the initial time and the norm of the new initial data are the same (as done

in the other proves we just use at the end of the estimates $\|\cdot\|_{[0,T]}$ instead of $\|\cdot\|_{[0,t]}$)

It follows everything as in the rough exponential.

The solution is actually in $D_X^{2\beta}$: we have to see that $Y, Y' \in C^\beta$ and $R^Y \in C^{2\beta}$:

$$\cdot |R_{s,t}^Y| = |Y_{s,t} - Y'_s X_{s,t}| = \left| \int_s^t f(Y_u) dX_u - f(Y_s) X_{s,t} \right|$$

$$\xrightarrow{\text{estimate in the def of rough int.}} \lesssim \|Df(Y)Y'\|_\infty |X_{s,t}| + O(|t-s|^{2\alpha})$$

$$\Rightarrow \|R_{s,t}^Y\|_{2\beta} \lesssim \|Df(Y)Y'\|_\infty \|X\|_{2\beta} \stackrel{X \in C^{2\beta}}{\leq} \infty$$

$$\cdot Y \in C^\beta: Y_{s,t} = Y'_s X_{s,t} + R_{s,t}^Y \Rightarrow \|Y\|_\beta \leq \|Y'\|_\infty \|X\|_\beta + \|R^Y\|_{2\beta} T^\beta \stackrel{X \in C^\beta}{\leq} \infty$$

$$\cdot Y' \in C^\beta: |Y'_{s,t}| = |f(Y)_{s,t}| \leq \|Df\|_\infty |Y_{s,t}| \Rightarrow \|Y'_{s,t}\|_\beta \lesssim \|Y\|_\beta$$

Also, since $D_X^{2\beta} \subset D_X^{2\alpha}$ it has to be unique in $D_X^{2\beta}$ as well.

□

Itô-Lyons map: given $f \in C_b^3([0,T]; \mathcal{L}(\mathbb{R}^m, \mathcal{L}(\mathbb{R}^d, \mathbb{R}^m)))$, the map

$$C^\beta([0,T]; \mathbb{R}^d) \times \mathbb{R}^m \longrightarrow D_X^{2\beta}([0,T]; \mathbb{R}^d)$$

$$(\mathbf{X}, y) \longmapsto (Y, Y') \text{ s.t. is the solution of}$$

$$Y_\cdot = y + \int_0^\cdot f(Y_s) d\mathbf{X}_s \text{ with } Y' = f(Y)$$

is the Itô-Lyons map.

Continuity of the Itô-Lyons map: let $\beta \in (\frac{1}{3}, \frac{1}{2}]$ and $f \in C_b^3$. Let $\mathbf{X} = (X, X)$,

$$\tilde{\mathbf{X}} = (\tilde{X}, \tilde{X}) \in C^\beta \text{ and } y, \tilde{y} \in \mathbb{R}^m. \text{ Let } (Y, Y') \in D_X^{2\beta} \text{ and } (\tilde{Y}, \tilde{Y}') \in D_{\tilde{X}}^{2\beta}$$

be the solutions of the RDE seen before with data $(y, \mathbf{X})$ and $(\tilde{y}, \tilde{\mathbf{X}})$.

Let $M > 0$ be a constant s.t. $\|\mathbf{X}\|_\beta \|\tilde{\mathbf{X}}\|_\beta \leq M$. Then, for any $\alpha \in (\frac{1}{3}, \beta) \exists C_M > 0$

constant depending on $\alpha, T, \|f\|_{C_b^3}$ and M s.t.

$$\|Y - \tilde{Y}\|_\alpha + \|Y' - \tilde{Y}'\|_\alpha + \|R^Y - R^{\tilde{Y}}\|_{2\alpha} \leq C_M (|y - \tilde{y}| + \|\mathbf{X}; \tilde{\mathbf{X}}\|_\alpha)$$

In particular the Itô-Lyons map is loc. lip. in a weaker topology ($\|\cdot\|_\alpha$ not $\|\cdot\|_\beta$).

↳ Proof: by the proof of the previous theorem $\exists t_2, \tilde{t}_2$ s.t. $\|Y, Y'\|_{X, \alpha, [0, t_2]} \leq 1$

and $\|\tilde{Y}, \tilde{Y}'\|_{X, \alpha, [0, \tilde{t}_2]} \leq 1$ (they are in B_{t_2} and $B_{\tilde{t}_2}$).

We use $\lesssim$ also if depend on M .

By the last lemma of the previous section $\forall t \in (t_2 \wedge \tilde{t}_2)$ (we can use it for these)

$$\|Y' - \tilde{Y}'\|_{\alpha; [0, t]} = \|f(Y) - f(\tilde{Y})\|_{\alpha; [0, t]}$$

$$\lesssim |Y_0 - \tilde{Y}_0| + (|Y_0' - \tilde{Y}_0'| + \|Y' - \tilde{Y}'\|_{\alpha; [0, t]}) \|X\|_{\alpha; [0, t]} + \|R^Y - R^{\tilde{Y}}\|_{2\alpha; [0, t]} t^\alpha + \|X - \tilde{X}\|_{\alpha; [0, t]}$$

$$\|R^Y - R^{\tilde{Y}}\|_{2\alpha; [0, t]} = \|R \int_0^t f(X_u) d\tilde{X}_u - R \int_0^t f(\tilde{X}_u) d\tilde{X}_u\|_{2\alpha; [0, t]} \lesssim$$

$$\lesssim (|Y_0 - \tilde{Y}_0| + |Y_0' - \tilde{Y}_0'| + \|Y' - \tilde{Y}'\|_{\alpha; [0, t]} + \|R^Y - R^{\tilde{Y}}\|_{2\alpha; [0, t]} + \|X - \tilde{X}\|_{\alpha; [0, t]}) \|\tilde{X}\|_{\alpha; [0, t]} + \|\tilde{X}, \tilde{X}\|_{\alpha; [0, t]}$$

$$\text{But } |Y_0 - \tilde{Y}_0| = |f(Y_0) - f(\tilde{Y}_0)| \stackrel{f \text{ lip.}}{\lesssim} |Y_0 - \tilde{Y}_0| \text{ so}$$

$$\|Y' - \tilde{Y}'\|_{\alpha; [0, t]} + \|R^Y - R^{\tilde{Y}}\|_{2\alpha; [0, t]} \leq C_3 (|Y_0 - \tilde{Y}_0| + (\|Y' - \tilde{Y}'\|_{\alpha; [0, t]} + \|R^Y - R^{\tilde{Y}}\|_{2\alpha; [0, t]})) (\|\tilde{X}\|_{\alpha; [0, t]} + t^\alpha + \|\tilde{X}, \tilde{X}\|_{\alpha})$$

$$\text{Now } \|\tilde{X}\|_{\alpha; [0, t]} + t^\alpha \leq \|X\|_{\beta; [0, t]} t^{\beta-\alpha} + \|X\|_{2\beta; [0, t]} t^{\frac{2(\beta-\alpha)}{2}} + t^\alpha$$

So choosing $t_3 \in (0, t_2 \wedge \tilde{t}_2)$ sufficiently small s.t.

$$C_3 \left(\|X\|_{\beta; T} t_3^{\beta-\alpha} + \|X\|_{2\beta; T} t_3^{\frac{2(\beta-\alpha)}{2}} + t_3^\alpha \right) \leq \frac{1}{2}$$

we find that

$$(*)_1 \quad \|Y' - \tilde{Y}'\|_{\alpha; [0, t_3]} + \|R^Y - R^{\tilde{Y}}\|_{2\alpha; [0, t_3]} \lesssim |Y_0 - \tilde{Y}_0| + \|\tilde{X}; \tilde{X}\|_{\alpha; [0, t_3]}$$

Using the estimates given in the rough int. theorem on $\|Y - \tilde{Y}\|_{\alpha}$:

$$(*)_2 \quad \|Y - \tilde{Y}\|_{\alpha; [0, t_3]} \lesssim |\tilde{Y}_0 - \tilde{Y}_0| + \|\tilde{X}; \tilde{X}\|_{\alpha; [0, t_3]}$$

So, we proceed exactly in the proof of the continuity of the solutions of Yang of E,

defining translation (as in the rough exp. proof) to obtain $(*)_1$ and $(*)_2$ for all $[s, t]$ s.t.

$|t-s| < \delta$ for a particular δ and following the iter of the proof. $\square$

Consistency with stochastic calculus

Consistency with stochastic integration

Consistency with $B^{It\hat{o}}$: let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \in [0, T]}, \mathbb{P})$ be a filtered probability space.

Let $\alpha \in (\frac{1}{3}, \frac{1}{2})$ and $B^{It\hat{o}} = (B, B^{It\hat{o}})$ be an $\mathcal{F}_t$ -adapted Itô enhanced Brownian rough path so that $B^{It\hat{o}} \in C^\alpha$ almost surely. Let (Y, Y') be an adapted stochastic process s.t. $(Y(\omega), Y'(\omega)) \in \mathcal{D}_{B(\omega)}^{2\alpha}$ for almost surely $\omega \in \Omega$. Then

$$\int_0^T Y_u d B_u^{It\hat{o}} = \int_0^T Y_u dB_u$$

almost surely. It means $\forall \omega \in \Omega$:

$$\int_0^T Y_u(\omega) d B_u^{It\hat{o}}(\omega) = \left(\int_0^T Y_u dB_u \right)(\omega)$$

where the LHS is defined as $\lim_{|\pi| \rightarrow 0} \sum_{[s, t] \in \pi} Y_s(\omega) B_{s, t}(\omega) + Y'_s(\omega) B_{s, t}^{It\hat{o}}(\omega)$.

→ Remark: Y adapted w.r.t. $\mathcal{F}_t$ + continuous $\Rightarrow Y$ predictable w.r.t. $\mathcal{F}_t$;

furthermore choosing $\tau_n := \inf \{ t \in [0, T] : |Y_t| \geq n \}$ $\wedge T$ is a stopping time
↖ Frobenius norm
↑ we use it in case the inf = $\emptyset$

$$\left(\{ \tau_n \leq t \} = \left\{ \sup_{u \leq t} |Y_u| \geq n \right\} \in \mathcal{F}_t \right. \quad \left. \uparrow \text{continuity of } Y \text{ (we can approximate with } \mathbb{Q}) \right)$$

and $\tau_n \uparrow T$ $\left(\forall \omega \quad \eta(\omega) := \sup_{t \leq T} |Y_t(\omega)| < \infty \right)$ continuous on a compact so $\forall n > \eta(\omega)$ we have

$\tau_n(\omega) = T$ so that

$$\int_0^T (Y_t^{\tau_n})^2 dt \leq \int_0^T n^2 dt = n^2 T < \infty$$

So $\int_0^T Y_u dB_u$ is a well defined local martingale given by the limit

$$\lim_{|\pi| \rightarrow 0} \sum_{[s, t] \in \pi} Y_s B_{s, t} \xrightarrow{\mathbb{P}} \int_0^T Y_u dB_u$$

↳ Proof: let $(\pi^n)_{n \geq 1}$ be a sequence of partition with $|\pi^n| \rightarrow 0$ as $n \rightarrow +\infty$ s.t.

$$(*)_1 \quad \sum_{[s,t] \in \pi^n} Y_s B_{s,t} \xrightarrow{P} \int_0^T Y_u dB_u \quad \text{as } n \rightarrow +\infty.$$

On the other hand, by definition of rough integral

$$\sum_{[s,t] \in \pi^n} (Y_s B_{s,t} + Y'_s \overset{It\hat{o}}{B}_{s,t}) \xrightarrow{a.s.} \int_0^T Y_u d\overset{It\hat{o}}{B}_u \quad \text{as } n \rightarrow +\infty.$$

So, in particular the convergence is in probability.

$$(*)_2 \quad \sum_{[s,t] \in \pi^n} (Y_s B_{s,t} + Y'_s \overset{It\hat{o}}{B}_{s,t}) \xrightarrow{P} \int_0^T Y_u d\overset{It\hat{o}}{B}_u \quad \text{as } n \rightarrow +\infty.$$

Now setting $A_n := \sum_{[s,t] \in \pi^n} Y_s B_{s,t}$, $B_n := \sum_{[s,t] \in \pi^n} Y'_s \overset{It\hat{o}}{B}_{s,t}$, $C_n := \sum_{[s,t] \in \pi^n} (Y_s B_{s,t} + Y'_s \overset{It\hat{o}}{B}_{s,t})$

since they are finite sum $A_n + B_n = C_n$ and $A_n \xrightarrow{P} \int_0^T Y_u dB_u$ $(*)_1$

$C_n \xrightarrow{P} \int_0^T Y_u d\overset{It\hat{o}}{B}_u$ $(*)_2$ so

$$B_n = C_n - A_n \xrightarrow{P} \int_0^T Y_u d\overset{It\hat{o}}{B}_u - \int_0^T Y_u dB_u$$

property of the limit in probability

If we see that $B_n \xrightarrow{P} 0$ then since the limit in prob. is a.s. unique we would have the result.

Step 1: property using orthogonality of martingale increments: $\forall \pi$ partition

$$\mathbb{E} \left[\left| \sum_{[s,t] \in \pi} Y'_s \overset{It\hat{o}}{B}_{s,t} \right|^2 \right] = \sum_{[s,t] \in \pi} \mathbb{E} \left[\left| Y'_s \overset{It\hat{o}}{B}_{s,t} \right|^2 \right] + 2 \sum_{\substack{[u,v], [s,t] \in \pi \\ v \leq s}} \underbrace{\mathbb{E} \left[\langle Y'_u \overset{It\hat{o}}{B}_{u,v}, Y'_s \overset{It\hat{o}}{B}_{s,t} \rangle \right]}_{(*)_3}$$

but $(*)_3 = \mathbb{E} \left[\mathbb{E} \left[\langle Y'_u \overset{It\hat{o}}{B}_{u,v}, Y'_s \overset{It\hat{o}}{B}_{s,t} \rangle \mid \mathcal{F}_s \right] \right]$

$$u, v, s \leq t \Rightarrow \mathbb{E} \left[\langle Y'_u \overset{It\hat{o}}{B}_{u,v}, Y'_s \mathbb{E} \left[\overset{It\hat{o}}{B}_{s,t} \mid \mathcal{F}_s \right] \rangle \right] = 0$$

$\int_0^t \overset{It\hat{o}}{B}_{u,v} d\overset{It\hat{o}}{B}_u$ is a martingale so the increments are zero

Case when $\|Y'\|_{L^\infty(\Omega \times [0,T])} \leq M$: $\forall \pi$ partition

$$\mathbb{E} \left[\sum_{[s,t] \in \pi} Y_s' B_{s,t}^{It\hat{\alpha}} \right]^2 = \sum_{[s,t] \in \pi} \mathbb{E} \left[|Y_s' B_{s,t}^{It\hat{\alpha}}|^2 \right]$$

previous step

hp. of this step $\rightarrow \leq K^2 \sum_{[s,t] \in \pi} \mathbb{E} [|B_{s,t}^{It\hat{\alpha}}|^2]$

proof that $B^{It\hat{\alpha}}$ is a rough path:

$$\mathbb{E} [|B_{s,t}^{It\hat{\alpha}}|^2] \leq C_4^2 |t-s|^2$$

for some constant $C_4^2 =: C$

So taking $\pi := \pi^n$ we have $\mathbb{E} [|B_n|^2] \xrightarrow{n \rightarrow +\infty} 0$ and so $B_n \xrightarrow{L^2} 0$, in particular $B_n \xrightarrow{P} 0$ as we wanted.

Case when $\|Y'\|_{L^\infty(\mathcal{L}_X[0,T])} = \infty$: let $\eta > 0$ and we define the stopping

time $\hat{\tau}_\eta = T \wedge \inf \{ t \in [0, T] : |Y_t'| \geq \eta \}$ (it is a stopping because Y_t' continuous, as in the remark for Y_t) and $\hat{\tau}_\eta \uparrow T$ (as in the remark for $\hat{\tau}_\eta$).

Now we use the same exact arguments for the stopped processes $(Y_{t \wedge \hat{\tau}_\eta}, Y'_{t \wedge \hat{\tau}_\eta})_t$

and $B_{\hat{\tau}_\eta}^{It\hat{\alpha}} := (B_t^{\hat{\tau}_\eta}, (B_{s,t}^{It\hat{\alpha}})^{\hat{\tau}_\eta})_t$ where

$$\begin{aligned} (B_{s,t}^{It\hat{\alpha}})^{\hat{\tau}_\eta} &\stackrel{\text{def}}{=} \int_s^t B_{s,r}^{\hat{\tau}_\eta} \otimes dB_r^{\hat{\tau}_\eta} \stackrel{\text{property of } It\hat{\alpha} \text{ integral}}{=} \int_s^{t \wedge \hat{\tau}_\eta} B_{s,r}^{\hat{\tau}_\eta} \otimes dB_r = \\ &\stackrel{\text{def.}}{=} \int_s^{t \wedge \hat{\tau}_\eta} (B_{r \wedge \hat{\tau}_\eta} - B_{s \wedge \hat{\tau}_\eta}) \otimes dB_r \end{aligned}$$

In particular, we can see that $B_{\hat{\tau}_\eta}^{It\hat{\alpha}}$ is still in $\mathcal{C}^\alpha$ a.s. and $(Y_{t \wedge \hat{\tau}_\eta}, Y'_{t \wedge \hat{\tau}_\eta})_t$

is in $\mathcal{D}_{B_{\hat{\tau}_\eta}^{\hat{\tau}_\eta}(w)}^{\mathcal{Z}\alpha}$ a.s. and so the first part still hold. Furthermore, for the

stopping time theorem $(B_{s,t}^{It\hat{\alpha}})^{\hat{\tau}_\eta}$ is still a martingale and so the increments are still

orthogonal. Also the estimate on $\mathbb{E} [|(B_{s,t}^{It\hat{\alpha}})^{\hat{\tau}_\eta}|^2] = O(|t-s|^2)$ holds (same proof

using BDG in the stopping time version). So, we find that $\forall \eta$

$$\int_0^T Y_{\hat{\tau}_\eta}^{\hat{\tau}_\eta} d(B^{It\hat{\alpha}})^{\hat{\tau}_\eta} = \int_0^T Y_{\hat{\tau}_\eta}^{\hat{\tau}_\eta} B_{\hat{\tau}_\eta}^{\hat{\tau}_\eta} \quad \text{a.s.}$$

but for the properties of rough integrals and $It\hat{\alpha}$ integrals, when they are defined,

$$\int_0^T Y^\pi d\tilde{X}^\pi = \int_0^{\tilde{T} \wedge T} Y d\tilde{X}^\pi \quad \text{and} \quad \int_0^T H^\pi dN^\pi = \int_0^{\tilde{T} \wedge T} H dN \quad (H \text{ predictable, } N \text{ semimartingale})$$

so

$$\int_0^{T \wedge \tilde{T}} Y_u d\tilde{B}_u^{\text{It}\hat{\sigma}} = \int_0^{T \wedge \tilde{T}} Y_u dB_u \quad \text{s.s.} \quad \text{and sending } T \rightarrow \infty, \tilde{T} \rightarrow T$$

and we have the result. $\square$

Consistency with B^{strat} : let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \in [0, T]}, \mathbb{P})$ be a filtered probability space.

let $\alpha \in (\frac{1}{3}, \frac{1}{2})$ and $B^{\text{strat}} = (B, B^{\text{It}\hat{\sigma}})$ be an $\mathcal{F}_t$ -adapted Stratonovich enhanced Brownian rough path so that $B^{\text{strat}} \in \mathcal{C}_g^{2\alpha}$ almost surely. Let (Y, Y') be an adapted stochastic process

s.t. $(Y(\omega), Y'(\omega)) \in \mathcal{D}_{B(\omega)}^{2\alpha}$ for almost surely $\omega \in \Omega$. Then

$$\int_0^T Y_u d\tilde{B}_u^{\text{strat}} = \int_0^T Y_u \circ dB_u$$

almost surely. It means $\forall \omega \in \Omega$:

$$\int_0^T Y_u(\omega) d\tilde{B}_u^{\text{strat}}(\omega) = \left(\int_0^T Y_u \circ dB_u \right)(\omega)$$

where the LHS is defined as $\lim_{|\pi| \rightarrow 0} \sum_{[s, t] \in \pi} Y_s(\omega) B_{s, t}(\omega) + Y'_s(\omega) \tilde{B}_{s, t}^{\text{strat}}(\omega)$.

Proof We know that:

$$\int_0^T Y_u d\tilde{B}_u^{\text{strat}} = \int_0^T Y_u d\tilde{B}_u^{\text{It}\hat{\sigma}} + \frac{1}{2} \int_0^T Y'_u du \quad \begin{matrix} \uparrow \\ \text{previous} \\ \text{prop s.s.} \end{matrix}$$

$$\tilde{B}_{s, t}^{\text{strat}} = \tilde{B}_{s, t}^{\text{It}\hat{\sigma}} + \frac{1}{2} (t-s)I$$

+ "Integral formula for rough path

with different second level" with $F_{s, t} = \frac{1}{2} (t-s)I$:

$$\text{we have to compute } \int Y'_u dF_u := \lim_{|\pi| \rightarrow 0} \sum_{[s, t] \in \pi} (Y'_s) \left(\frac{1}{2} (t-s)I \right) = \lim_{|\pi| \rightarrow 0} \sum_{[s, t] \in \pi} \frac{1}{2} (t-s) Y'_s(I) =: \int_0^T Y'_u du \quad \in \mathbb{R}^m$$

So if we see that $\langle Y, B \rangle_T = \int_0^T Y'_u du$, we have finished since $\forall M, N$ semimartingale:

$$\int_0^T H_t \circ dN_t = \int_0^T H_t dN_t + \frac{1}{2} \langle M, N \rangle_t.$$

hence, focus on $\langle Y, B \rangle_T$:

$$\langle Y, B \rangle_T \stackrel{\text{def.}}{=} \lim_{|\pi| \rightarrow 0}^P \sum_{[s,t] \in \pi} Y_{s,t} B_{s,t} \quad \leftarrow \text{in probability}$$

$$Y_{s,t} = Y_s^1 B_{s,t} + R_{s,t}^Y \rightarrow \lim_{|\pi| \rightarrow 0}^P \sum_{[s,t] \in \pi} (Y_s^1 B_{s,t} + R_{s,t}^Y) B_{s,t}$$

$$(Y_s^1 B_{s,t}) B_{s,t} = \lim_{|\pi| \rightarrow 0}^P \sum_{[s,t] \in \pi} Y_s^1 (B_{s,t} \otimes B_{s,t}) + R_{s,t}^Y B_{s,t}$$

$$= Y_s^1 (B_{s,t} \otimes B_{s,t})$$

using the ident.

$$d(\mathbb{R}^d, d(\mathbb{R}^d, \mathbb{R}^m)) \simeq d(\mathbb{R}^d \otimes \mathbb{R}^d; \mathbb{R}^n)$$

$$\begin{aligned} \frac{1}{2} B_{s,t} \otimes B_{s,t} &= \lim_{|\pi| \rightarrow 0}^P \sum_{[s,t] \in \pi} \frac{1}{2} Y_s^1 \text{Sym}(B_{s,t}^{It\hat{\alpha}}) + (t-s) Y_s^1(I) + R_{s,t}^Y B_{s,t} \\ &= \text{Sym}(B_{s,t}^{It\hat{\alpha}}) + \frac{1}{2}(t-s)I \end{aligned}$$

So, we have 3 terms:

$$\lim_{|\pi| \rightarrow 0}^P \sum_{[s,t] \in \pi} \frac{1}{2} Y_s^1 \text{Sym}(B_{s,t}^{It\hat{\alpha}}) : \text{Sym is a linear operator so } B_{s,t}^{It\hat{\alpha}} \text{ is still}$$

a martingale and $|\text{Sym}(B_{s,t}^{It\hat{\alpha}})| \leq |B_{s,t}^{It\hat{\alpha}}|$ so using the same argument of the previous

proof, $\sum_{[s,t] \in \pi} Y_s^1 \text{Sym}(B_{s,t}^{It\hat{\alpha}}) \xrightarrow{L^2} 0$; so this term is 0.

$$\lim_{|\pi| \rightarrow 0} \sum_{[s,t] \in \pi} (t-s) Y_s^1(I) = \int_0^T Y_u^1 du \text{ by definition (the def. is a.s. } \Rightarrow P)$$

$$\lim_{|\pi| \rightarrow 0} \sum_{[s,t] \in \pi} R_{s,t}^Y B_{s,t} : \left| \sum_{[s,t] \in \pi} R_{s,t}^Y B_{s,t} \right| \leq \|R^Y\|_{2\alpha} \|B\|_{\alpha} \sum_{[s,t] \in \pi} |t-s|^{3\alpha}$$

$$\leq \|R^Y\|_{2\alpha} \|B\|_{\alpha} T |\pi|^{3\alpha-1} \xrightarrow{|\pi| \rightarrow 0} 0 \quad (\text{q.e. } \Rightarrow P)$$

So we have obtained what we wanted.

□

Consistency with SDE

Consistency of RDE with SDE: let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \in [0,T]}, \mathbb{P})$ be a filtered probability

space. Let $f \in C_b^3$, $\alpha \in (\frac{1}{3}, \frac{1}{2})$ and $g \in L^2(\mathbb{P})$. Let B be a d -dimensional BM.

(i) If $B^{It\hat{\sigma}} = (B, \bar{B}^{It\hat{\sigma}}) \in \mathcal{C}^\alpha$ a.s. is $It\hat{\sigma}$ enhanced Brownian rough path, and

$(y(\omega), y'(\omega))$ is the solution of the RDE

$$dY_t(\omega) = f(Y_t(\omega)) d\hat{B}^{\tau_t \hat{\omega}}(\omega), \quad Y_0 = y$$

then $(Y_t)_{t \in [0, T]}$ is the unique strong solution of the Itô SDE:

$$dY_t = f(Y_t) dB_t, \quad Y_0 = y$$

(i) If $B^{\text{Strat}} = (B, B^{\text{Itô}}) \in \mathcal{C}_g^{0,\alpha}$ as is Stratonovich enhanced Brownian rough path, and

$(Y(\omega), Y'(\omega))$ is the solution of the RDE

$$dY_t(\omega) = f(Y_t(\omega)) d\mathbf{B}^{\text{Strat}}(\omega), \quad Y_0 = y$$

then $(Y_t)_{t \in [0, T]}$ is the unique strong solution of the Itô SDE:

$$dy_t = f(y_t) \circ dB_t, \quad y_0 = y$$

Proof Basically, to use the previous two propositions we have to see that $f(y)$ is adapted

to y_t , but we see more: $\forall$ is B_t -adapted and so the solution is also a strong

solution (it is more because $\sigma(B_t) \subseteq \mathcal{F}_t$ since B_t is $\mathcal{F}_t$ -adapted and since f is

deterministic obviously $f(Y_t) \subseteq \sigma(B_t) \subseteq \mathcal{Y}_t$. It is the only thing to verify since if it is true:

we omit $\longrightarrow$ the dependence on ω here

$$Y_t = y + \int_0^t f(Y_u) d\hat{B}_u^{\text{Itô/Strat}} = y + \int_0^t f(Y_u) d\hat{B}_u / \int_0^t f(Y_u) \circ d\hat{B}_u$$

$\uparrow$
def. of sol.

$\uparrow$
previous propositions

But $\sigma(B_s, B_{s,r}^{\pm t_0} : 0 \leq s \leq r \leq t) = \sigma(B_u : 0 \leq u \leq t)$ (the same for Strt. since they

$$\hat{B}_{s,r}^{\text{Itô}} = \int_s^t B_{r,s} \circ B_s$$

differ from a deterministic factor)

and so fixed $t \in [a, T]$:

$$\leftarrow (\beta_s, \|\beta_{s,v} : s \leq v \leq t)$$

$$(\Omega, \mathcal{F}, \mathbb{P}) \ni \omega \longmapsto (B(\omega)|_{[0,t]}, \mathbb{B}(\omega)|_{[0,t]}) \in \mathcal{C}^{\mathbb{B}}([0,t])$$

this is important: to construct $Y|_{[a,t]}$ we $\longrightarrow$ only need the path between $[a,t]$

$$\left(\gamma(w) \Big|_{[0,t]}, \gamma'(w) \Big|_{[0,t]} \right) \in \mathcal{D}_{B(w)}^{2\beta}([0,t])$$

Since $\Phi_{[s,t]}(\omega)$ is continuous Y is a well defined stoch. process B_t -adapted.

Pathwise stability of likelihood estimators

The classical MLE

Setting: $\mathbb{V}$ finite-dimensional vector space, let $\Sigma \in \mathbb{R}^{d \times d}$ and let $h: \mathbb{R}^d \rightarrow \mathbb{L}(\mathbb{V}; \mathbb{R}^d)$

be a Lipschitz continuous map. Consider the stochastic dynamics:

$$dX_t = h(X_t) A dt + \Sigma dW_t^A, \quad X_0 = x_0$$

where $A \in \mathbb{V}$ is a parameter, W^A d-dim. $\mathbb{P}^A$ -B.M. and $x_0 \in \mathbb{R}^d$ is the deterministic initial value.
 it varies $\downarrow$ *they depend on A because we only count that is a B.M.*

We assume Σ is invertible s.t. $C := \Sigma \Sigma^T$ is a positive definite symmetric matrix

Aim: observing X up to some T , we want to estimate the parameter A , and so we consider

the Maximum Likelihood Estimator (MLE) $\hat{A}_T$ defined as a map:

$$\hat{A}_T: C([0, T], \mathbb{R}^d) \rightarrow \mathbb{V}$$

the path observed for the realization ω $\rightarrow (X(\omega))_{t \in [0, T]} \mapsto \hat{A}_T(X(\omega))$ *the corresponding "best" estimate*

What is the "best" A ? : if we think that under a certain prob $\mathbb{P}^0$

$$dX_t = \Sigma dW_t^0, \quad X_0 = x_0$$

we can define

$$W_t^A(\omega) := W_t(\omega) - \int_0^t f_s^A(\omega) ds \quad \text{with} \quad f_t^A(\omega) := \Sigma^{-1} h(X_t(\omega)) A \in \mathbb{R}^d$$

and by Girsanov theorem W_t^A is a $\mathbb{P}^A$ -B.M. where $\mathbb{P}^A$ is defined by the density

w.r.t. $\mathbb{P}^0$:

$$\begin{aligned} L(A) &:= \frac{d\mathbb{P}^A}{d\mathbb{P}^0} = \exp \left(\int_0^T (f_s^A)^T dW_s^0 - \frac{1}{2} \int_0^T |f_s^A|^2 ds \right) \\ &= \exp \left(\int_0^T \overset{\text{in the coordinate sense}}{A^T h(X_s)^T} \Sigma^{-T} dW_s^0 - \frac{1}{2} \int_0^T A^T h(X_s)^T \Sigma^{-T} \Sigma^{-1} h(X_s) A ds \right) \\ &= \exp \left(\int_0^T A^T h(X_s)^T C^{-1} dX_s - \frac{1}{2} \int_0^T A^T h(X_s)^T C^{-1} h(X_s) A ds \right) \end{aligned}$$

the integral is linear

$$\Rightarrow = \exp \left(A^T \int_0^T h(X_s)^T C^{-1} dX_s - \frac{1}{2} A^T \int_0^T h(X_s)^T C^{-1} h(X_s) ds A \right)$$

$$=: \exp \left(A^T S_T - \frac{1}{2} A^T I_T A \right)$$

In particular the model becomes what we wanted:

$$\begin{aligned} dX_t &= \sum dW_t^a = \sum \sum^{-1} h(X_t) A dt + \sum dW_t^A \\ &= h(X_t) A dt + \sum dW_t^A \end{aligned}$$

In the RLE "the best" A is the one that maximizes:

$$L(A)(\omega) \quad \text{w.r.t. } A \quad (\omega \text{ is fixed, in the sense that } (X_t(\omega))_{t \in [0, T]} \text{ is given})$$

So, the best estimates $\hat{A}$ satisfy:

$$\left. \frac{dL(A)(\omega)}{dA} \right|_{A=\hat{A}} = 0$$

hence, it's simpler to compute $\frac{d}{dA} \log L(A)(\omega)$

$$\begin{aligned} \frac{d}{dA} \log L(A)(\omega) &= \frac{d}{dA} \left(A^T S_T(\omega) - \frac{1}{2} A^T I_T(\omega) A \right) \\ &= S_T(\omega) - I_T(\omega) A = 0 \end{aligned}$$

Hence $I_T(\omega) A = S_T(\omega)$ where (underlying the dependence on ω)

$$S_T := \int_0^T h(X_s)^T C^{-1} dX_s \in V^* \quad (\text{in coordinates we have indeed } A \mapsto A^T S_T)$$

$$I_T := \int_0^T h(X_s)^T C^{-1} h(X_s) ds \in \mathcal{L}(V, V^*) \quad (\text{in coordinates } A \mapsto I_T A : B \mapsto B^T I_T A)$$

So, if I_T is invertible:

$$A = I_T^{-1} S_T.$$

I_T is invertible: in the setting above we define

$$R_h := \{X \in C([0, T]; \mathbb{R}^d) : \forall A \in V \text{ with } A \neq 0, \exists t \in T \text{ s.t. } h(X_t) A \neq 0\}$$

Then, if $\mathbb{P}^0(X \in R_h) = 1 \Rightarrow I_T^{-1}$ is defined a.s.

↓
Proof In general $f \in \mathcal{L}(V, V^*)$ is invertible $\Leftrightarrow \text{Ker } f = 0$ since we are in finite dimensional space; so f is invertible $\Leftrightarrow$ the only element $v \in V$ s.t. $f(v)(w) = 0 \quad \forall w$ is $v = 0$;
 but if f is symmetric and positive semidefinite it is sufficient to see

$$f(v)(v) = 0 \Rightarrow v = 0$$

since for Cauchy Schwarz for symm. positive semidef. f holds $|f(v)(w)| \leq \sqrt{f(v,v)f(w,w)}$.

Now I_T is obviously symm. and pos. sem. $\left(A^T I_T A = \int_0^T \| C^{-\frac{1}{2}} h(X_s) A \|^2 ds \geq 0 \right)$

Now

C pos. def. and see

$C^{-\frac{1}{2}}$ exists because C is positive definite!

$$A^T I_T A = 0 \Rightarrow h(X_s) A = 0 \quad \forall s \in [0, T] \Rightarrow X \notin R_h.$$

□

Lack of continuity for the classical MLE

Let us start with an example where we have continuity w.r.t. the supremum norm (it works basically because we will work in dimension 1):

Example where there is continuity: W, X 1-dim., $W = \mathbb{R}$, $\Sigma = \sigma > 0$ and h the identity map on $\mathbb{R}$ so that

$$dX_t = A X_t dt + \sigma dW_t^A, \quad X_0 = x_0 \neq 0 \text{ (under } \mathbb{P}^A)$$

In this case:

$$I_T = \int_0^T h(X_s) C^{-1} h(X_s) ds = \int_0^T X_s \sigma^{-2} X_s ds = \sigma^{-2} \int_0^T X_s^2 ds$$

↑
are numbers

$$S_T = \int_0^T h(X_s) C^{-1} dX_s = \sigma^{-2} \int_0^T X_s dX_s = \frac{\sigma^{-2}}{2} \left(X_T^2 - x_0^2 - \langle X \rangle_T \right)$$

↑
Itô formula + $\langle X \rangle_0 = 0$

$$X_t = \int_0^t \sigma dW_s \Rightarrow \frac{\sigma^{-2}}{2} \left(X_T^2 - x_0^2 - \sigma^2 T \right)$$

$$\text{Now } \hat{A}_T = I_T^{-1} S_T = \frac{1}{2} \frac{X_T^2 - x_0^2 - \sigma^2 T}{\int_0^T X_s^2 ds} \quad \leftarrow \text{this is not zero since } X_0 = x_0 \neq 0 + \text{continuity of } X$$

So, fixing ω , we choose $X(\omega), X(\tilde{\omega}) \in C([0, T], \mathbb{R})$ s.t. $X_0(\omega) = X_0(\tilde{\omega})$ then

it is clear that if $|\hat{A}_T(\omega) - \hat{A}_T(\tilde{\omega})|$ is locally controlled by $\|X(\omega) - X(\tilde{\omega})\|_\infty$

$$(\text{the leb. integral}) : \left| \int_0^T X_s^2(\omega) - X_s^2(\tilde{\omega}) ds \right| \leq \int_0^T |X_s(\omega) + X_s(\tilde{\omega})| |X_s(\omega) - X_s(\tilde{\omega})| ds$$

$$\leq T (\|X\|_\infty + \|\tilde{X}\|_\infty) (\|X(\omega) - X(\tilde{\omega})\|_\infty)$$

Example with lack of continuity: W and $X = (X_1, X_2)$ are 2-dim., $V = \mathbb{R}^{2 \times 2}$,

$\Sigma = \text{Id}_{2 \times 2}$, $h: \mathbb{R}^2 \rightarrow \mathcal{L}(\mathbb{R}^{2 \times 2}, \mathbb{R}^2)$ s.t. $h(v)(A) = Av$, so the underlying dynamics are

$$dX_t = h(X_t)A dt + dW_t = AX_t dt + dW_t$$

Now:

$$S_T(H) = \int_0^T (h(X_s)H)^T dX_s = \int_0^T X_s^T H^T dX_s \stackrel{\text{the argument is a scalar}}{=} \int_0^T \text{tr}(X_s^T H^T dX_s)$$

$$= \int_0^T \text{tr}(H^T dX_s X_s^T) \stackrel{\text{linearity}}{=} \text{tr}\left(H^T \int_0^T dX_s X_s^T\right)$$

$$\mathbb{I}_T(A, H) = \int_0^T (h(X_s)H)^T (h(X_s)A) ds = \int_0^T (HX_s)^T (AX_s) ds =$$

$$= \int_0^T X_s^T H^T A X_s ds \stackrel{\text{the argument is a scalar}}{=} \int_0^T \text{tr}(X_s^T H^T A X_s) ds$$

$$\text{tr}(ABC) = \text{tr}(CAB) \rightarrow \int_0^T \text{tr}(H^T A X_s X_s^T) ds = \text{tr}(BCA) \text{ (cycle)}$$

$$\text{linearity of the integral} \rightarrow \text{tr}\left(H^T A \int_0^T X_s X_s^T ds\right) =$$

$$\text{So } \mathbb{I}_T(A, \cdot) = S_T(\cdot) \Leftrightarrow A \int_0^T X_s X_s^T ds = \int_0^T dX_s X_s^T \Leftrightarrow A = \left(\int_0^T dX_s X_s^T\right) \left(\int_0^T X_s X_s^T ds\right)^{-1}$$

Now:

$$\left(\int_0^T dX_s X_s^T\right) = \begin{pmatrix} \int_0^T X_s^1 dX_s^1 & \int_0^T X_s^2 dX_s^1 \\ \int_0^T X_s^1 dX_s^2 & \int_0^T X_s^2 dX_s^2 \end{pmatrix}$$

, the same for $\left(\int_0^T X_s X_s^T ds\right)$ and we know for a 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ the

inverse is $\frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ so

$$\left(\int_0^T X_s X_s^T ds\right)^{-1} = \frac{1}{\underbrace{\int_0^T (X_s^1)^2 ds \int_0^T (X_s^2)^2 ds - \left(\int_0^T X_s^1 X_s^2 ds\right)^2}_{\Delta_T}} \begin{pmatrix} \int_0^T (X_s^2)^2 ds & -\int_0^T X_s^1 X_s^2 ds \\ -\int_0^T X_s^1 X_s^2 ds & \int_0^T (X_s^1)^2 ds \end{pmatrix}$$

So :

$$A^{1,1} = \frac{1}{U_T} \left(\int_0^T (X_s^1)^2 ds \int_0^T X_s^1 dX_s^1 - \int_0^T X_s^1 X_s^2 ds \int_0^T X_s^2 dX_s^1 \right)$$

Now all the terms that are Lebesgue integral are continuous w.r.t. supremum norm, also

$$(X_T^1)^2 = (X_0^1)^2 + 2 \int_0^T X_s^1 dX_s^1 + T \Rightarrow \int_0^T X_s^1 dX_s^1 = \frac{1}{2} ((X_T^1)^2 - (X_0^1)^2 - T)$$
 so it is continuous

in the supremum norm, instead $\int_0^T X_s^2 dX_s^1$ will not in the sense of the map:

$$\Phi : (C([0, T]; \mathbb{R}^2), \|\cdot\|_\infty) \longrightarrow \mathbb{R}$$

$$(X_s(\omega))_{s \in [0, T]} \longmapsto \left(\int_0^T X_s^2 dX_s^1 \right)(\omega)$$

as ω varies,
the path varies as well

this is in the Itô sense

Now choosing :

$$X : [0, 1] \times \Omega \longrightarrow \mathbb{R}^2 \text{ s.t. for a fixed } \bar{\omega}$$

→ this example does not give an $X = x_0 + B$ but it shows anyway that there could be some problems

$X(\bar{\omega}) : [0, 1] \longrightarrow \mathbb{R}^2$ has the support in the square $(0, 0), (1, 0), (1, 1)$ and $(0, 1)$

at constant speed, anticlockwise and starting and finishing in $(0, 0)$. We have

$$\int_0^1 X_s^2 dX_s^1(\bar{\omega}) = \int_0^1 X_s^2(\bar{\omega}) dX_s^1(\bar{\omega}) = \int_{\text{right}} + \int_{\text{top}} + \int_{\text{left}} + \int_{\text{bottom}} =$$

the trajectories has finite variation $\Rightarrow$ the integral is pathwise

$$= 0 + 0 + 1(-1) + 0 = -1.$$

Then, we choose $(\omega_n)_{n \in \mathbb{N}}$ s.t. $X_t(\omega_n)$ is as $X(\bar{\omega})$ but at the end we put

a "fast loop", we define $X_t(\omega_n)$:

• square: $X_t(\omega_n) := X_{\frac{n}{n-1}}(\bar{\omega})t, t \in [0, \frac{n-1}{n}]$ ← in this way we come back after we have done all the square

• fast loop $X_t(\omega_n) := \left(\frac{1}{n} (\cos \theta_t - 1), \frac{1}{n} \sin \theta_t \right), \theta_t := 2\pi n^2 \left(t - \frac{n-1}{n} \right) t \in \left[\frac{n-1}{n}, 1 \right]$

So $\int_0^1 X_s^2 dX_s^1(w_n) = \underbrace{\int_0^{\frac{n-1}{n}} X_{\frac{n}{n-1}}^2(\bar{w}) dX_{\frac{n}{n-1}}^1(\bar{w})}_{\substack{\text{the same integral as above} \\ = -1}} + \underbrace{\int_{\frac{n-1}{n}}^1 X_t^2(w_n) dX_t^1(w_n)}_{\substack{\text{you are doing a circle of area } \pi(\frac{1}{n})^2 \text{ anticlock} \\ \text{with } \theta = \omega T \quad \omega = 2\pi n^5 \text{ and } T = \frac{1}{n} \\ \Rightarrow \text{you are doing it } n^3. \text{ So:} \\ = -\pi(\frac{1}{n})^2 n^3 = -\pi n}}$

as before the trajectories are of finite variation

Hence $\|X(\bar{w}) - X(w_n)\|_\infty \xrightarrow[\substack{\text{the area of} \\ \text{the circle goes to zero}}]{\text{the area of}} 0$ but the value of $\int_0^1 X_s^2 dX_s^1(w_n) \xrightarrow{n \rightarrow \infty} -\infty$.

MLE via rough paths

We need a stronger topology than the uniform one to restore stability.

We remember that:

- $dX_t = h(X_t) A dt + \sum dW_t^A, \quad X_0 = x_0 \quad (\text{under } \mathbb{P}^A)$
- $X_t = x_0 + \sum W_t \quad \text{under } \mathbb{P}^0$

We require more regularity for h , indeed we take $h \in C_b^2(\mathbb{R}^d; \mathcal{L}(V; \mathbb{R}^d))$.

Lift X_t to a rough path: we define the second level of X_t as

$$X_{s,t}^{(2)} := \left(\int_s^t X_{s,r} \otimes dX_r \right)(\omega) \quad \forall \omega \in \Omega \quad \forall (s,t) \in \Delta_{[0,T]}$$

← the Itô's integral under $\mathbb{P}^0$!

Then, $X = (X, X^{(2)})$ is almost surely an α -holder rough path for any $\alpha \in (\frac{1}{3}, \frac{1}{2})$.

→ Proof: Now

$$\begin{aligned} X_{s,t}^{(2)} &= \int_s^t \sum (W_r - W_s) \otimes \sum dW_r = \sum \left(\int_s^t W_r - W_s \otimes dW_r \right) \sum^T \\ X_t &= x_0 + \sum W_t = \sum W_{s,t} \sum^T \end{aligned}$$

So $\|X_{s,t}\|_{L^2} \leq \|\sum\|^2 \|W_{s,t}\|_{L^2} \Rightarrow$ using the estimate for $\beta = \frac{1}{2}$ used

in the proof to see the brownian motion can be lifted to a rough path using the Kolmogorov criterion we obtain the result.

□

Remark: the lift $X_{s,t}$ defined above is almost surely the same integral if we use an equivalent measure $\mathbb{P}^A$, indeed:

$$\sum_{[u,v] \in \pi_n} X_{s,u} \otimes X_{u,v} \xrightarrow[\|\pi_n\| \rightarrow 0]{\mathbb{P}} \int_s^t X_{s,r} \otimes dX_r^{(\mathbb{P})}$$

$\downarrow$

$$\exists (\pi_n)_n \text{ s.t. } \sum_{[u,v] \in \pi_n} (X_{s,u} \otimes X_{u,v})(\omega) \longrightarrow \int_s^t X_{s,r} \otimes dX_r^{(\mathbb{P})}(\omega) \quad \mathbb{P}\text{-a.s.}$$

but $\mathbb{P} \sim \mathbb{P}^A$ so this limit holds $\mathbb{P}^A$ -a.s. as well but then, since it always

hold by definition $\sum_{[u,v] \in \pi_n} (X_{s,u} \otimes X_{u,v}) \xrightarrow{\mathbb{P}^A} \int_s^t X_{s,r} \otimes dX_r^{(\mathbb{P}^A)}$ we must have

$$\int_s^t X_{s,r} \otimes dX_r^{(\mathbb{P})} = \int_s^t X_{s,r} \otimes dX_r^{(\mathbb{P}^A)} \quad \forall \omega \quad (\text{since the uniqueness of the limit in prob.})$$

Setting: we can define

$$\mathbb{D} := \{ (X, X) \in \mathcal{C}^\alpha : X \in R_h \}$$

So that if $\mathbb{P}^0(X \in R_h) = 1 \Rightarrow \mathbb{P}^0(X \in \mathbb{D}) = 1$

• the bilinear form I_T which we can consider as a map $R_h \rightarrow \mathcal{L}(V, V^*)$ does not change since the integral used is already pathwise:

$$I_T(X) = \int_0^T h(X_s)^T C^{-1} h(X_s) ds$$

• we define $S_T(X, X) := \int_0^T h(X_s)^T C^{-1} dX_s$ (now it is a pathwise integral) and since $h \in C_b^2$ it is well defined.

Robust MLE: we define the robust MLE $\hat{A}_T: \mathbb{D} \rightarrow V$ given by

$$\hat{A}_T(X, X) = I_T^{-1}(X) S_T(X, X) \quad \leftarrow \text{always evaluated in "w"}$$

The robust MLE is continuous: the map $\hat{A}_T$ defined above is a continuous map from $\mathbb{D}$ to

V w.r.t. the α -Hölder rough path distance. Moreover, writing $X = (X, X)$ for the

If $\hat{\tau}$ rough path lift of X as above, we have:

$$\mathbb{P}^\circ(\hat{A}_T(\hat{X}) = \hat{A}_T(X)) = 1$$

→ Proof:

Continuity: I_T is clearly continuous w.r.t. the weaker supremum norm and hence so its inverse (which is defined for all $X \in \mathbb{D}$). The map S_T is continuous w.r.t. the rough path distance given in the prop. "continuity of the rough integral". The composition of α -holder is α -holder.

Equality: it's the consistency of the stochastic integral with the rough integral (we have shown it for the BIt but it is the same since X is a linear transformation of a brownian motion with a drift term, so the extension is trivial).

□

The stochastic sewing lemma

Problem: suppose we have B a brownian motion and $\bar{X} = (X, \mathbb{X}) \in \mathcal{C}^\alpha$ with $\alpha \in (\frac{1}{2}, \frac{1}{2}]$.

Then, if we want to define $\int_0^t f(B_r + X_r) d\bar{X}_r$ we could try:

• stochastic integration: X has not necessarily bounded variation so it is not a semimartingale $\Rightarrow$ we

can't use stochastic integration

↑
1-Hölder $\Rightarrow$ BV
but here $\alpha \in (\frac{1}{2}, \frac{1}{2}]$

• rough integration: we would require that $f(B_r + X_r)$ is almost surely controlled by $\bar{X}$,

but fixed ω :

$$f(B_t(\omega) + X_t) \stackrel{\uparrow}{=} f(B_s(\omega) + X_s) + f'(B_s(\omega) + X_s)(B_{s,t}(\omega) + X_{s,t}) + f''(\xi_{s,t}(\omega))(B_{s,t}(\omega) + X_{s,t})^2$$

Taylor expansion:

$$\phi(z_t) = \phi(z_s) + \phi'(z_s)(z_t - z_s) + \phi''(\xi_{s,t})(z_t - z_s)^2 \text{ for some } \xi_{s,t}$$

$$\Rightarrow f(B(\omega) + X)_{s,t} = f'(B_s(\omega) + X_s) X_{s,t} + f'(B_s(\omega) + X_s) B_{s,t}(\omega) + f''(\xi_{s,t}(\omega))(B_{s,t}(\omega) + X_{s,t})^2$$

so it is natural to set the Gubinelli derivative $= f'(B_s(\omega) + X_s)$ and the remainder

$$R_{s,t}(\omega) := f'(B_s(\omega) + X_s) B_{s,t}(\omega) + f''(\xi_{s,t}(\omega))(B_{s,t}(\omega) + X_{s,t})^2$$

but the remainder is not 2α -Hölder regular.

So, both stochastic and rough approaches can't handle such an integral. We need another sewing lemma.

The stochastic sewing lemma

Usual estimate: $(\Omega, \mathcal{F}, P)$ prob. space $Z_1, \dots, Z_n$ a finite sequence of $\mathbb{R}^d$ -valued random variables

in $L^m(\Omega, \mathcal{F}, P)$ for some $m \in [2, +\infty)$. Suppose that we are given a filtration $(\mathcal{F}_i)_{i=1, \dots, n}$

s.t. for each i , the variables $Z_1, \dots, Z_{i-1}$ are all $\mathcal{F}_i$ -measurable. Then, set $S := \sum_{i=1}^n Z_i$

it holds:

$$\|S\|_{L^m} \leq \sum_{i=1}^n \|\mathbb{E}[Z_i | \mathcal{F}_i]\|_{L^m} + \underbrace{C_m}_{\text{BDG constant}} \left(\sum_{i=1}^n \|Z_i\|_{L^m}^2 \right)^{\frac{1}{2}}$$

Proof: we can rewrite:

$$S = S_1 + S_2 \quad \text{where} \quad S_1 := \sum_{i=1}^n \mathbb{E}[z_i | \mathcal{Y}_i] \quad \text{and} \quad S_2 := \sum_{i=1}^n (z_i - \mathbb{E}[z_i | \mathcal{Y}_i])$$

So:

$$\|S\|_{L^1} \leq \|S_1\|_{L^m} + \|S_2\|_{L^m} \quad \text{and}$$

$$\cdot \|S_1\|_{L^m} \leq \sum_{i=1}^n \|\mathbb{E}[z_i | \mathcal{Y}_i]\|_{L^m}$$

$$\cdot \|S_2\|_{L^m} = \mathbb{E} \left[\left| \sum_{i=1}^n (z_i - \mathbb{E}[z_i | \mathcal{Y}_i]) \right|^m \right]^{\frac{1}{m}} \leq$$

BHG inequality in discrete case state that given a martingale X_n difference $dn := X_n - X_{n-1}$ then

$\exists C_p$ s.t. $\mathbb{E}[|X_n|^m] \leq C_p \mathbb{E} \left[\left(\sum_{i=1}^n d_n^2 \right)^{\frac{m}{2}} \right]$, use $X_j := \sum_{i=1}^j (z_i - \mathbb{E}[z_i | \mathcal{Y}_i])$

$$\leq C_m \mathbb{E} \left[\left(\sum_{i=1}^n |z_i - \mathbb{E}[z_i | \mathcal{Y}_i]|^2 \right)^{\frac{m}{2}} \right]^{\frac{1}{m}}$$

$$= C_m \left\| \sum_{i=1}^n |z_i - \mathbb{E}[z_i | \mathcal{Y}_i]|^2 \right\|_{L^{m/2}}^{\frac{1}{2}}$$

Δ inequality of the $L^{m/2}$ -norm $\rightarrow \leq C_m \left(\sum_{i=1}^n \| |z_i - \mathbb{E}[z_i | \mathcal{Y}_i]|^2 \|_{L^{m/2}} \right)^{\frac{1}{2}}$

$$= C_m \left(\sum_{i=1}^n \|z_i - \mathbb{E}[z_i | \mathcal{Y}_i]\|_{L^m}^2 \right)^{\frac{1}{2}}$$

$$\leq C_m \left(\sum_{i=1}^n (\|z_i\|_{L^m}^2 + \|\mathbb{E}[z_i | \mathcal{Y}_i]\|_{L^m}^2) \right)^{\frac{1}{2}}$$

$$\mathbb{E} \left[\mathbb{E}[z_i | \mathcal{Y}_i]^m \right]^{\frac{1}{m}} \leq C_m \left(\sum_{i=1}^n \|z_i\|_{L^m}^2 \right)^{\frac{1}{2}}$$

$$\stackrel{\uparrow \text{Jensen}}{\leq} \mathbb{E} \left[\mathbb{E}[z_i^m | \mathcal{Y}_i] \right]^{\frac{1}{m}} = \mathbb{E} \left[z_i^m \right]^{\frac{1}{m}}$$

□

Stochastic sewing Lemma: let $(\mathcal{R}, \mathcal{F}, (\mathcal{Y}_t)_{t \in [0, T]}, \mathbb{P})$ be a filtered prob. space, and let

$m \geq 2$. Let $A: \Delta_{[0, T]} \rightarrow L^m$ be a continuous map, such that, $\forall s \leq t$ $A_{s, s} = 0$

and $A_{s, t}$ is $\mathcal{Y}_t$ -adapted. Suppose that, for some constants $\lambda_1, \lambda_2 \geq 0$ and $\epsilon_1, \epsilon_2 > 0$, we

have:

$$\cdot \| \mathbb{E}[\delta A_{s, u, t} | \mathcal{Y}_s] \|_{L^m} \leq \lambda_1 |t-s|^{1+\epsilon_1}$$

$$\cdot \| \delta A_{s, u, t} \|_{L^m} \leq \lambda_2 |t-s|^{\frac{1}{2}+\epsilon_2}$$

$\forall 0 \leq s \leq u \leq t \leq T$.

Then, $\exists!$ (up to modification) stochastic process $\gamma = (\gamma_t)_{t \in [0, T]}$ s.t.

(i) $\gamma_0 = 0$ and γ is $\mathcal{F}_t$ -adapted and L^m -integrable

(ii) $\exists C_1, C_2$ constants s.t.

$$\|\gamma_t - \gamma_s - A_{s,t}\|_{L^m} \leq C_1 |t-s|^{1+\varepsilon_1} + C_2 |t-s|^{\frac{1}{2}+\varepsilon_2}$$

$$\|\mathbb{E}[\gamma_t - \gamma_s - A_{s,t} | \mathcal{F}_s]\|_{L^m} \leq C_1 |t-s|^{1+\varepsilon_1} \quad \forall s, t \in \Delta_{[0, T]}.$$

Moreover, the constants C_1, C_2 may be taken to be $\lambda_1 (1-2^{-\varepsilon_1})^{-1}$ and $2 C_m \lambda_2 (1-2^{-\varepsilon_2})^{-1}$

resp. and $\forall (s, t) \in \Delta_{[0, T]}$ we have

↑
from the
previous lemma

$$\lim_{|\pi| \rightarrow 0} \sum_{[u, v] \in \pi} A_{u, v} = \gamma_t - \gamma_s$$

where the limit exists in L^m and is taken over any partition π of $[s, t]$ s.t. $|\pi| \rightarrow 0$.

Proof Step 1:

Fixed $[s, t] \subset [0, T]$. For each $n \geq 1$, let $\pi^n = \{s = t_0^n < t_1^n < \dots < t_{2^n}^n = t\}$ be the dyadic partition of $[s, t]$ with $|\pi^n| = \frac{1}{2^n} (t-s)$.

We define $\forall n \geq 1$:

$$A_{s,t}^n := \sum_{i=0}^{2^n-1} A_{t_i^n, t_{i+1}^n}$$

We want to see that this sequence is Cauchy in L^m : choosing u_i^n the midpoint of $[t_i^n, t_{i+1}^n]$

$$A_{s,t}^n - A_{s,t}^{n+1} = \sum_{i=0}^{2^n-1} \delta A_{t_i^n, u_i^n, t_{i+1}^n}, \quad \text{and choosing } \mathcal{F}_i := \mathcal{F}_{t_i^n, u_i^n, t_{i+1}^n} \text{ we can apply}$$

the previous lemma ($\mathcal{F}_i := \mathcal{F}_{t_i^n}$):

$$\|A_{s,t}^n - A_{s,t}^{n+1}\|_{L^m} \leq \sum_{i=0}^{2^n-1} \|\mathbb{E}[\delta A_{t_i^n, u_i^n, t_{i+1}^n} | \mathcal{F}_{t_i^n}]\|_{L^m} + 2 C_m \left(\sum_{i=0}^{2^n-1} \|\delta A_{t_i^n, u_i^n, t_{i+1}^n}\|_{L^m}^2 \right)^{\frac{1}{2}}$$

$$\text{estimates in hyp.} \Rightarrow \leq \lambda_1 \sum_{i=0}^{2^n-1} |t_{i+1}^n - t_i^n|^{1+\varepsilon_1} + 2 C_m \lambda_2 \left(\sum_{i=0}^{2^n-1} |t_{i+1}^n - t_i^n|^{1+2\varepsilon_2} \right)^{\frac{1}{2}}$$

$$|t_{i+1}^n - t_i^n| = \frac{1}{2^n} (t-s) \Rightarrow = \lambda_1 |t-s|^{1+\varepsilon_1} \sum_{i=0}^{2^n-1} 2^{-n(1+\varepsilon_1)} + 2 C_m \lambda_2 |t-s|^{\frac{1}{2}+\varepsilon_2} \left(\sum_{i=0}^{2^n-1} 2^{-n(1+2\varepsilon_2)} \right)^{\frac{1}{2}}$$

$$= \lambda_1 |t-s|^{1+\varepsilon_1} 2^{-n\varepsilon_1} + 2 C_m \lambda_2 |t-s|^{\frac{1}{2}+\varepsilon_2} 2^{-n\varepsilon_2} \xrightarrow{n \rightarrow \infty} 0$$

So it is Cauchy and exists the limit in L^m :

$$\lim_{n \rightarrow \infty} A_{s,t}^n =: \mathcal{I}_{s,t}$$

and satisfies

↖ see the proof of the sewing lemma

$$\|\mathcal{I}_{s,t} - A_{s,t}\|_{L^m} \leq$$

↗ this will be the 1st estimate we want to prove

$$\leq \sum_{n=0}^{+\infty} \|A_{s,t}^n - A_{s,t}^{n+1}\|_{L^m}$$

$$\leq \frac{\lambda_1}{1-z^{-\varepsilon_1}} |t-s|^{1+\varepsilon_1} + \frac{2C_m \lambda_2}{1-z^{-\varepsilon_2}} |t-s|^{\frac{1}{2}+\varepsilon_2}$$

$$\|\mathbb{E}[\mathcal{I}_{s,t} - A_{s,t} | \mathcal{Y}_s]\|_{L^m} \leq \sum_{n=0}^{+\infty} \|\mathbb{E}[A_{s,t}^n - A_{s,t}^{n+1} | \mathcal{Y}_s]\|_{L^m}$$

↗ this will be the 2nd estimate we want to prove

$$= \sum_{n=0}^{+\infty} \left\| \mathbb{E} \left[\sum_{i=0}^{2^n-1} \delta A_{t_i^n, u_i^n, t_{i+1}^n} \mid \mathcal{Y}_s \right] \right\|_{L^m}$$

$$\mathcal{Y}_s \subseteq \mathcal{Y}_{t_i^n} \rightarrow = \sum_{n=0}^{+\infty} \left\| \mathbb{E} \left[\mathbb{E} \left[\sum_{i=0}^{2^n-1} \delta A_{t_i^n, u_i^n, t_{i+1}^n} \mid \mathcal{Y}_{t_i^n} \right] \mid \mathcal{Y}_s \right] \right\|_{L^m}$$

as in the last step of the previous proof ↖

$$\leq \sum_{n=0}^{+\infty} \left\| \mathbb{E} \left[\sum_{i=0}^{2^n-1} \delta A_{t_i^n, u_i^n, t_{i+1}^n} \mid \mathcal{Y}_{t_i^n} \right] \right\|_{L^m}$$

as before ↖

$$\leq \frac{\lambda_1}{1-z^{-\varepsilon_1}} |t-s|^{1+\varepsilon_1}$$

Step 2: existence of γ : A^n is continuous as function from $\Delta_{[0,T]} \rightarrow L^m$ and the

convergence $A_{s,t}^n \rightarrow \mathcal{I}_{s,t}$ is uniform in the L^m -norm $\Rightarrow \mathcal{I}$ is itself continuous.

We define $\gamma_t := \mathcal{I}_{0,t}$ and as in the proof of the "Sewing lemma" we see the additivity,

$\gamma_0 = 0$; then, γ_t is $\mathcal{Y}_t$ -measurable and in L^m (obvious) (i), and rewriting the previous estimates with γ we obtain the point (ii). (the previous estimates hold $\forall (s,t)$, we use the dyadic only to approximate).

Step 3: uniqueness: let $\bar{\gamma}$ another $\mathcal{Y}_t$ adapted process satisfying (i) and (ii). We define

$$\zeta = \gamma - \bar{\gamma}. \text{ Then}$$

$$(\textcircled{e}_1) \quad \|\zeta_t - \zeta_s\|_{L^m} \leq \tilde{C} |t-s|^{\frac{1}{2} + \tilde{\varepsilon}}$$

$$(\textcircled{e}_2) \quad \|\mathbb{E}[\zeta_t - \zeta_s | \mathcal{Y}_s]\|_{L^m} \leq \tilde{C} |t-s|^{1+\tilde{\varepsilon}} \quad \forall (s,t) \in \Delta_{[0,T]}, \text{ for some } \tilde{C}, \tilde{\varepsilon} > 0.$$

We use the dyadic partition $\{t_i\}$ so that by telescopic sum

$$Z_t = \sum_{i=0}^{2^n-1} (Z_{t_{i+1}} - Z_{t_i}) \quad (Z_0 = \gamma_0 - \bar{\gamma}_0 = 0)$$

we can apply the previous lemma and the estimates $(\mathcal{Q}_1), (\mathcal{Q}_2)$ and as in the 1st step

$$\text{we see that } \|Z_t\|_{L^m} \lesssim \sum_{i=0}^{2^n-1} 2^{-n(1+\tilde{\varepsilon})} + \left(\sum_{i=0}^{2^n-1} 2^{-n(1+2\tilde{\varepsilon})} \right)^{\frac{1}{2}} \lesssim 2^{-n\tilde{\varepsilon}}$$

$$\Rightarrow \|Z_t\|_{L^m} = 0 \Rightarrow \gamma_t = \bar{\gamma}_t \text{ a.s.}$$

Step 4 : convergence of Riemann sums : we take any partition $\pi = \{t_i\}_{i=0}^N$ of $[s, t]$, then

$$\|\gamma_t - \gamma_s - \sum_{i=0}^{N-1} A_{t_i, t_{i+1}}\|_{L^m} =$$

$$\gamma_t - \gamma_s = \sum_{i=1}^{N-1} \gamma_{t_{i+1}} - \gamma_{t_i} \rightarrow = \left\| \sum_{i=0}^{N-1} \gamma_{t_{i+1}} - \gamma_{t_i} - A_{t_i, t_{i+1}} \right\|_{L^m}$$

$$\xrightarrow{\text{previous lemma}} \leq \sum_{i=0}^{2^n-1} \|\mathbb{E}[\gamma_{t_{i+1}} - \gamma_{t_i} - A_{t_i, t_{i+1}} | \mathcal{F}_{t_i}]\|_{L^m} +$$

$$+ 2 C_m \left(\sum_{i=0}^{2^n-1} \|\gamma_{t_{i+1}} - \gamma_{t_i} - A_{t_i, t_{i+1}}\|_{L^m}^2 \right)^{\frac{1}{2}}$$

$$\xrightarrow{\text{estimates given by (ii)}} \lesssim \sum_{i=0}^{N-1} |t_{i+1} - t_i|^{1+\varepsilon_1} + \left(\sum_{i=0}^{N-1} |t_{i+1} - t_i|^{2+2\varepsilon_1} + |t_{i+1} - t_i|^{1+2\varepsilon_2} \right)^{\frac{1}{2}}$$

$$\lesssim |\pi|^{\varepsilon_1} + |\pi|^{\frac{1}{2}+\varepsilon_1} + |\pi|^{\varepsilon_2} \xrightarrow{|\pi| \rightarrow 0} 0.$$

□

Relation to Itô calculus

The stochastic sewing lemma : let B a standard BM in $\mathbb{R}^d$ w.r.t. the filtration $(\mathcal{F}_t)_{t \in [0, T]}$.

gives us the Itô integration

Suppose $f: \mathbb{R}^d \rightarrow L(\mathbb{R}^d; \mathbb{R})$ is β -Hölder continuous for some

$\beta \in [0, 1]$. Then, using the stochastic sewing lemma we have that it is well defined:

$$\int_0^t f(B_r) dB_r := \lim_{|\pi| \rightarrow 0} \sum_{[u, v] \in \pi} f(B_u) B_{u, v}$$

as limit in L^m . Moreover :

$$\left\| \int_s^t f(B_r) dB_r - f(B_s) B_{s, t} \right\|_{L^m} \lesssim |t-s|^{\frac{1}{2} + \beta_2}$$

$$\cdot \mathbb{E} \left[\int_s^t f(B_r) dB_r \mid \mathcal{F}_s \right] = 0 \quad (\text{in particular } \int_0^\cdot f(B_r) dB_r \text{ is a martingale})$$

$\cdot \forall \varepsilon > 0, \int_0^\cdot f(B_r) dB_r$ is the unique $\mathcal{F}_t$ -adapted L^m -integrable process γ with $\gamma_0 = 0$ s.t.

$$\|\gamma_t - \gamma_s - f(B_s) B_{s,t}\|_{L^m} \lesssim |t-s|^{1+\varepsilon} + |t-s|^{\frac{1}{2} + \frac{p}{2}}$$

$$\left(\|\mathbb{E}[\gamma_t - \gamma_s \mid \mathcal{F}_s]\|_{L^m} \lesssim |t-s|^{1+\varepsilon} \quad \forall (s,t) \in \Delta_{[0,T]} \right) \text{ Penso sia sbagliato}$$

→ Proof: $A_{s,t} := f(B_s) B_{s,t}$. Then $\delta A_{s,u,t} = f(B_s) B_{s,t} - f(B_u) B_{u,t} - f(B_s) B_{s,u}$
 $= -f(B)_{s,u} B_{u,t}$.

Now

$$\cdot \mathbb{E}[\delta A_{s,u,t} \mid \mathcal{F}_s] = -\mathbb{E}[f(B)_{s,u} B_{u,t} \mid \mathcal{F}_s] \stackrel{f(B)_{s,u} \text{ is } \mathcal{F}_u\text{-meas}}{=} 0$$

$$= -\mathbb{E}\left[f(B)_{s,u} \underbrace{\mathbb{E}[B_{u,t} \mid \mathcal{F}_u]}_{\substack{= \mathbb{E}[B_{u,t}] = 0 \\ B_{u,t} \text{ } \mathcal{F}_u\text{-meas.}}} \mid \mathcal{F}_s\right] = 0$$

$B_{u,t} \sim N(0, |t-u|I)$

$$\cdot \|\delta A_{s,u,t}\|_{L^m} \stackrel{f(B)_{s,u} \text{ is ind. from } B_{u,t}}{=} \|f(B)_{s,u}\|_{L^m} \|B_{u,t}\|_{L^m}$$

$f(B)_{s,u}$ is ind. from $B_{u,t}$ and for ind. r.v. $\|XY\|_{L^m} = \|X\|_{L^m} \|Y\|_{L^m}$

$$|f(B_u) - f(B_s)| \leq \|f\|_{C_b^p} \|B_{s,u}\|_{L^{pm}}^p \|B_{u,t}\|_{L^m}$$

$$\|f\|_{C_b^p} \|B_u - B_s\|^p$$

$$\begin{aligned} B_{s,u} = B_1(u-s) &\rightarrow \|f\|_{C_b^p} \|B_1\|_{L^{pm}}^p |u-s|^{\frac{p}{2}} \|B_1\|_{L^m} |t-u|^{\frac{1}{2}} \\ &\lesssim |t-s|^{\frac{p}{2} + \frac{1}{2}} \end{aligned}$$

So we can use the stochastic sewing lemma with $\lambda_1 = 0$ and $\varepsilon_2 = \frac{p}{2}$. □

The stochastic sewing lemma: let π be a L^2 -integrable $\mathcal{F}_t$ -adapted martingale in $\mathbb{R}^d$, and

and the quadratic covariation

assume: $\|\pi_{s,u} \otimes \pi_{u,t}\|_{L^2} \leq C |t-s|^{\frac{1}{2} + \varepsilon}$

for all $s \leq u \leq t$ for some constants C and ε . (For example

for the B.M. this condition is satisfied with $\varepsilon = \frac{1}{2}$.

$$\mathbb{E}[(B_u - B_s)^2 (B_t - B_u)^2]^{\frac{1}{2}} \stackrel{\text{ind. incr.}}{=} \mathbb{E}[(B_u - B_s)^2]^{\frac{1}{2}} \mathbb{E}[(B_t - B_u)^2]^{\frac{1}{2}}$$

Then, $\exists!$ $\mathcal{H}_t$ -adapted process $\langle \mathcal{M} \rangle$ s.t. $\langle \mathcal{M} \rangle_0 = 0$ and s.t.

$$= \mathbb{E}[(B_u - B_s)^2]^{\frac{1}{2}} \mathbb{E}[(B_t - B_u)^2]^{\frac{1}{2}} \\ \stackrel{N(0, u-s)}{=} \sqrt{u-s} \stackrel{N(0, t-u)}{=} \sqrt{t-u} \leq \sqrt{u-s} \sqrt{t-u} \stackrel{\substack{\uparrow \\ a \cdot b \leq \frac{a^2 + b^2}{2}}}{\leq} \frac{u-s+t-u}{2} = \frac{t-s}{2}$$

$$\|\langle \mathcal{M} \rangle_{s,t} - \mathcal{M}_{s,t} \otimes \mathcal{M}_{s,t}\|_{L^2} \lesssim |t-s|^{\frac{1}{2} + \varepsilon}$$

and $\mathbb{E}[\langle \mathcal{M} \rangle_{s,t} | \mathcal{H}_s] = \mathbb{E}[\mathcal{M}_{s,t} \otimes \mathcal{M}_{s,t}]$. Moreover we have

$$\langle \mathcal{M} \rangle_t = \lim_{|\pi| \rightarrow 0} \sum_{(u,v] \in \pi} \mathcal{M}_{u,v} \otimes \mathcal{M}_{u,v} \quad (\text{in } L^2)$$

Also, $\mathcal{M} \otimes \mathcal{M} - \langle \mathcal{M} \rangle$ is a martingale.

→ Proof: $A_{s,t} := \mathcal{M}_{s,t} \otimes \mathcal{M}_{s,t}$. Then

$$\begin{aligned} \delta A_{s,u,t} &= \mathcal{M}_{s,t} \otimes \mathcal{M}_{s,t} - \mathcal{M}_{s,u} \otimes \mathcal{M}_{s,u} - \mathcal{M}_{u,t} \otimes \mathcal{M}_{u,t} \\ &\stackrel{\mathcal{M}_{s,t} = \mathcal{M}_{s,u} + \mathcal{M}_{u,t}}{=} (\mathcal{M}_{s,u} + \mathcal{M}_{u,t}) \otimes (\mathcal{M}_{s,u} + \mathcal{M}_{u,t}) - \mathcal{M}_{s,u} \otimes \mathcal{M}_{s,u} - \mathcal{M}_{u,t} \otimes \mathcal{M}_{u,t} \\ &= \mathcal{M}_{s,u} \otimes \mathcal{M}_{u,t} + \mathcal{M}_{u,t} \otimes \mathcal{M}_{s,u} \end{aligned}$$

So using the estimate in the hyp:

$$\|\delta A_{s,u,t}\|_{L^2} \leq 2C |t-s|^{\frac{1}{2} + \varepsilon}$$

and also we have

$$\mathbb{E}[\delta A_{s,u,t} | \mathcal{H}_s] \stackrel{\substack{\uparrow \\ \text{orthogonal increments for a martingale}}}{=} 0$$

So the existence and the properties of $\langle \mathcal{M} \rangle$ are given by the stochastic sewing lemma.

The fact that $\mathcal{M} \otimes \mathcal{M} - \langle \mathcal{M} \rangle$ is a martingale follows by:

$$\mathbb{E}[\mathcal{M}_t \otimes \mathcal{M}_t - \langle \mathcal{M} \rangle_t - (\mathcal{M}_s \otimes \mathcal{M}_s - \langle \mathcal{M} \rangle_s) | \mathcal{H}_s] =$$

$$\mathbb{E}[\mathcal{M}_t \otimes \mathcal{M}_t - \mathcal{M}_s \otimes \mathcal{M}_s | \mathcal{H}_s]$$

$$\mathbb{E}[\mathcal{M}_{s,t} \otimes \mathcal{M}_{s,t} | \mathcal{H}_s] \text{ because}$$

use that $\mathbb{E}[\mathcal{M}_t \otimes \mathcal{M}_s | \mathcal{H}_s] = \mathcal{M}_s \otimes \mathcal{M}_s$ since $\mathcal{M}_t$ mart. and $\mathcal{M}_s - \mathcal{H}_s$ -meas.

$$\Rightarrow \mathbb{E}[\mathcal{M}_{s,t} \otimes \mathcal{M}_{s,t} - \langle \mathcal{M} \rangle_{s,t} | \mathcal{H}_s] =$$

$$= \mathbb{E}[\mathcal{M}_{s,t} \otimes \mathcal{M}_{s,t} - \mathcal{M}_{s,t} \otimes \mathcal{M}_{s,t} | \mathcal{H}_s]$$

$$\mathbb{E}[\langle \mathcal{M} \rangle_{s,t} | \mathcal{H}_s] = \mathbb{E}[\mathcal{M}_{s,t} \otimes \mathcal{M}_{s,t}] = 0.$$

□

A rough stochastic integral

Now, we answer to the question given at the beginning of the chapter.

Rough stochastic integral for : let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \in [0, T]}, \mathbb{P})$ be a filtered prob. space and

a particular integrand $\alpha \in (\frac{1}{3}, \frac{1}{2}]$. Let $\mathbf{X} = (X, \mathbb{X}) \in \mathcal{C}^\alpha$ be a rough path and let

B be an $\mathcal{F}_t$ -adapted Brownian motion. Let $m \geq 2$ and $f \in C_b^2$. Then the limit

$$\int_0^t f(B_r + X_r) d\mathbf{X}_r := \lim_{|\Pi| \rightarrow 0} \sum_{[u, v] \in \Pi} f(B_u + X_u) X_{u, v} + Df(B_u + X_u) \mathbb{X}_{u, v}$$

exists in L^m $\forall t \in [0, T]$. Moreover, $\int_0^\cdot f(B_r + X_r) d\mathbf{X}_r$ is the unique $\mathcal{F}_t$ -adapted L^m -integrable process, started at zero, such that the estimates

$$\left\| \int_s^t f(B_r + X_r) d\mathbf{X}_r - f(B_s + X_s) X_{s, t} - Df(B_s + X_s) \mathbb{X}_{s, t} \right\|_{L^m} \lesssim |t-s|^{2\alpha}$$

$$\left\| \mathbb{E} \left[\int_s^t f(B_r + X_r) d\mathbf{X}_r - f(B_s + X_s) X_{s, t} - Df(B_s + X_s) \mathbb{X}_{s, t} \middle| \mathcal{F}_s \right] \right\|_{L^m} \lesssim |t-s|^{3\alpha}$$

hold for all $(s, t) \in \Delta_{[0, T]}$.

→ Proof : $A_{s, t} := f(B_s + X_s) X_{s, t} + Df(B_s + X_s) \mathbb{X}_{s, t}$

$$\begin{aligned} dA_{s, u, t} &= f(B_s + X_s) X_{s, t} + Df(B_s + X_s) \mathbb{X}_{s, t} \\ &\quad - f(B_s + X_s) X_{s, u} - Df(B_s + X_s) \mathbb{X}_{s, u} - f(B_u + X_u) X_{u, t} - Df(B_u + X_u) \mathbb{X}_{u, t} \end{aligned}$$

$$= f(B_s + X_s) X_{u, t} - f(B_u + X_u) X_{u, t} + Df(B_s + X_s) (X_{s, t} - X_{s, u}) - Df(B_u + X_u) \mathbb{X}_{u, t}$$

$$\text{Chain} \rightarrow = -f(B + X)_{s, u} X_{u, t} + Df(B_s + X_s) (X_{u, t} + X_{s, u} \otimes X_{u, t}) - Df(B_u + X_u) \mathbb{X}_{u, t}$$

$$= - (f(B + X)_{s, u} - Df(B_s + X_s) X_{s, u}) X_{u, t} - Df(B + X)_{s, u} \mathbb{X}_{u, t}$$

Now:

$$\begin{aligned} |f(x_1 - x_2)| &\leq \|f\|_{C_b^1} |x_1 - x_2| \\ \|f(B + X)_{s, u}\|_{L^m} &\leq \|f\|_{C_b^1} \|B_{s, u} + X_{s, u}\|_{L^m} \end{aligned}$$

$$\leq \|f\|_{C_b^1} (\|B_{s, u}\|_{L^m} + \|X_{s, u}\|_{L^m})$$

$$\leq \|f\|_{C_b^1} (\|B_1\|_{L^m} |u-s|^{\frac{1}{2}} + \|X\|_\alpha |u-s|^\alpha) \lesssim |t-s|^\alpha$$

wlog: $|\Pi| \rightarrow 0$

$\alpha \leq \frac{1}{2}$ $\wedge$ $|t-s| < 1$

Since $f \in C_b^2$, the same holds for $\|Df(B+X)_{s,u}\|_{L^m} \lesssim |t-s|^\alpha$

So

$$\|\delta A_{s,u,t}\|_{L^m} \lesssim (|t-s|^\alpha + |t-s|^\alpha |t-s|^\alpha) |t-s|^\alpha + |t-s|^\alpha |t-s|^{2\alpha} \lesssim |t-s|^{2\alpha}$$

Since $2\alpha > \frac{2}{3} > \frac{1}{2}$ we have the 1st inequality for the stoch. sewing lemma.

For the second one:

$$f(B+X)_{s,u} - Df(B_s+X_s)X_{s,u} =$$

classical
step +
sum and minus
 $Df(B_s+X_s)B_{s,u}$

$$= \int_0^1 (Df(B_s+X_s + r(B_{s,u}+X_{s,u})) - Df(B_s+X_s)) (B_{s,u}+X_{s,u}) dr + Df(B_s+X_s)B_{s,u}$$

Now taking $\mathbb{E}[\cdot | \mathcal{F}_s]$:

this term goes to zero with $\mathbb{E}[\cdot | \mathcal{F}_s]$
because of the orth. incr.

$$\begin{aligned} \|\mathbb{E}[f(B+X)_{s,u} - Df(B_s+X_s)X_{s,u}]\|_{L^m} &\lesssim \|f\|_{C_b^2} \|B_{s,u}+X_{s,u}\|^2_{L^m} \\ &\lesssim \|B_{s,u}\|_{L^{2m}}^2 + \|X_{s,u}\|^2 \\ &\lesssim \|B_u\|_{L^{2m}}^2 |u-s| + \|X\|_\alpha^2 |u-s|^{2\alpha} \\ &\lesssim |t-s|^{2\alpha} \end{aligned}$$

Hence

$$\|\mathbb{E}[\delta A_{s,u,t} | \mathcal{F}_s]\|_{L^m} \lesssim |t-s|^{3\alpha}, \text{ since } 3\alpha > 1 \text{ we have the II estimate needed}$$

to use the stoch. sewing lemma.

□

Rough paths of lower regularity

The signature and Chen's relation

We take $X = (X^1, \dots, X^d) : [0, T] \rightarrow \mathbb{R}^d$ smooth path.

Word: given $d \in \mathbb{N} \setminus \{0\}$, a word w of length $|w| = n > 0$ is

$$w = w_1 \dots w_n \quad \text{where } w_i \in \{1, \dots, d\} \quad \forall i = 1, \dots, n.$$

Given X as above, we now that its enhancement is the matrix X given by

$$(X)_{s,t}^{w_1 w_2} = \int_s^t X_{s,r_1}^{w_1} dX_{r_1,r_2}^{w_2} = \int_s^t \int_s^{r_2} dX_{r_1}^{w_1} dX_{r_2}^{w_2}, \quad w_1 w_2 \text{ word in the } \{1, \dots, d\} \text{ alphabet}$$

$$X_{r_2}^{w_1} - X_s^{w_1} = \int_s^{r_2} dX_{r_3}^{w_1}$$

So it seems natural this definition:

Iterated integrals: given $X : [0, T] \rightarrow \mathbb{R}^d$ smooth path, we can define the iterated integrals corresponding to a word w of length n :

$$X_{s,t}^w := \int_{\Delta_{[s,t]}^n} dX_{r_1}^{w_1} \dots dX_{r_n}^{w_n} \quad \text{where } \Delta_{[s,t]}^n := \{s < r_1 < \dots < r_n < t\}$$

↖ the n -dimensional simplex

$$= \int_s^t \int_s^{r_n} \dots \int_s^{r_3} \int_s^{r_2} dX_{r_1}^{w_1} dX_{r_2}^{w_2} \dots dX_{r_{n-1}}^{w_{n-1}} dX_{r_n}^{w_n}$$

Family of iterated integrals: given the words of length n , we have

$$\int_{\Delta_{[s,t]}^n} dX_{r_1} \otimes \dots \otimes dX_{r_n} \in (\mathbb{R}^d)^{\otimes n} \quad \text{in the sense}$$

$$\text{that } \left(\int_{\Delta_{[s,t]}^n} dX_{r_1} \otimes \dots \otimes dX_{r_n} \right)^w := \int_{\Delta_{[s,t]}^n} dX_{r_1}^{w_1} \dots dX_{r_n}^{w_n} \in \mathbb{R}$$

Signature of a smooth path : given $X: [0, T] \rightarrow \mathbb{R}^d$, we can define the signature of X over the interval $[s, t]$ as:

$$S(X)_{s,t} := \left(1, \int_s^t dX, \int_{\Delta_{[s,t]}^2} dX \otimes dX, \dots, \int_{\Delta_{[s,t]}^n} dX \otimes \dots \otimes dX, \dots \right) \in T(\mathbb{R}^d) = \bigoplus_{n=0}^{+\infty} (\mathbb{R}^d)^{\otimes n}$$

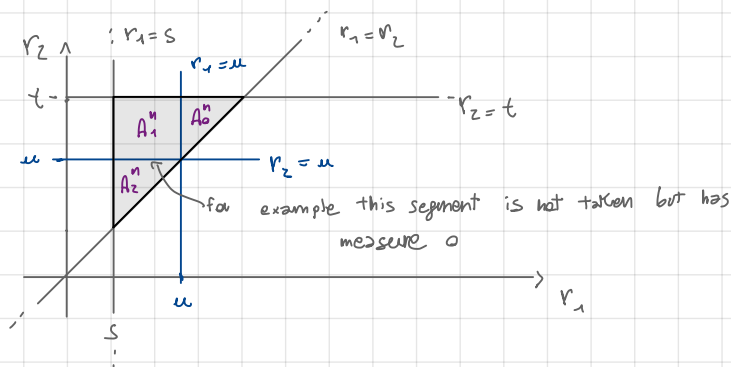
Lemma (decomposition of $\Delta_{[s,t]}^n$): let $s \leq u \leq t$, ignoring the set of zero n -dimensional Lebesgue measure, we have that:

$$\Delta_{[s,t]}^n = A_0^n \cup A_1^n \cup \dots \cup A_n^n$$

where $\forall 0 \leq j \leq n$:

$$A_j^n := \{ s < r_1 < \dots < r_j < u < r_{j+1} < \dots < r_n < t \} \stackrel{\text{obvious}}{=} \Delta_{[s,u]}^j \times \Delta_{[u,t]}^{n-j}$$

→ Geometric interpretation: for example we take $\Delta_{[s,t]}^2 = \{ (r_1, r_2) : s < r_1 < r_2 < t \}$, we cut with



→ Proof : given $(r_1, \dots, r_n) \in \Delta_{[s,t]}^n$, we fix $j = \max \{ k : r_k < u \} \in \{0, \dots, n\}$. ↙ convention $j=0$ if $r_1 \geq u$

Then $(r_1, \dots, r_n) \in A_j$ for some j unless $r_k = u$ for some k (by def A_j cannot have $r_k = u$ for some k) but it is an hyperplane $\Rightarrow$ Lebesgue measure n -dim. equals to zero.

Obviously the intersection $A_{j_1} \cap A_{j_2} = \emptyset$ if $j_1 \neq j_2$.

□

Decomposition of iterated integrals : given a word w of length $|w| = n \geq 1$, and given $s \leq u \leq t$:

$$X_{s,t}^w = \sum_{j=0}^n X_{s,u}^{w_1 \dots w_j} X_{u,t}^{w_{j+1} \dots w_n}$$

→ Proof $X_{s,t}^\omega = \int_{\Delta_{[s,t]}^n} dX_{r_1}^{\omega_1} \dots dX_{r_n}^{\omega_n} =$

previous decomposition $\rightarrow = \sum_{j=0}^n \int_{A_j^n} dX_{r_1}^{\omega_1} \dots dX_{r_n}^{\omega_n} =$

$$A_j^n = \Delta_{[s,u]}^j \times \Delta_{[u,t]}^{n-j} = \sum_{j=0}^n \left(\int_{\Delta_{[s,u]}^j} dX_{r_1}^{\omega_1} \dots dX_{r_j}^{\omega_j} \int_{\Delta_{[u,t]}^{n-j}} dX_{r_{j+1}}^{\omega_{j+1}} \dots dX_{r_n}^{\omega_n} \right)$$

$$= \sum_{j=0}^n X_{s,u}^{\omega_1 \dots \omega_j} X_{u,t}^{\omega_{j+1} \dots \omega_n}.$$

Truncated tensor algebra : $T^{(N)}(\mathbb{R}^d) = \bigoplus_{k=0}^N (\mathbb{R}^d)^{\otimes k}$, $S^N(X)_{s,t} \in T^{(N)}(\mathbb{R}^d)$ (truncated signature)

Product on the truncated tensor algebra : $|w| := N > 0$, $s < u < t$, we take

$$S^N(X)_{s,u} = \left(1, \int_s^u dX, \int_{\Delta_{[s,u]}^2} dX \otimes dX, \dots, \int_{\Delta_{[s,u]}^N} dX \otimes \dots \otimes dX \right) \in T^{(N)}(\mathbb{R}^d)$$

$$S^N(X)_{u,t} = \left(1, \int_u^t dX, \int_{\Delta_{[u,t]}^2} dX \otimes dX, \dots, \int_{\Delta_{[u,t]}^N} dX \otimes \dots \otimes dX \right) \in T^{(N)}(\mathbb{R}^d)$$

Then, we can define $S^N(X)_{s,u} * S^N(X)_{u,t} \in T^{(N)}(\mathbb{R}^d)$: in fact for every fixed level $j = 0, \dots, N$ and for every word $\omega_1 \dots \omega_j$ of length j we

$$\left(S^N(X)_{s,u} * S^N(X)_{u,t} \right)^{\omega_1 \dots \omega_j} := \sum_{i=0}^j X_{s,u}^{\omega_1 \dots \omega_i} X_{u,t}^{\omega_{i+1} \dots \omega_j}$$

$\leftarrow |w|=0 \Rightarrow X^\emptyset = 1$

Chen's relation for smooth paths : in the previous setting, using the decomposition of iterated integrals:

$$S^N(X)_{s,u} * S^N(X)_{u,t} = S^N(X)_{s,t}$$

Remark : obviously we can extend all to $S(X)_{s,t}$

• if we read the Chen's relation for smooth paths with in the third level ($|w|=2$),

we have:

$$X_{s,t}^{ij} = X_{u,t}^{ij} + X_{s,u}^i X_{u,t}^j + X_{s,u}^{ij}, \text{ hence}$$

$$\int_s^t X_{s,t_1}^i dX_{t_1}^j = \int_s^u X_{s,t_1}^i dX_{t_1}^j + \int_u^t X_{s,t_1}^i dX_{t_1}^j + X_{s,u}^i X_{u,t}^j$$

that is exactly the Chen's relation in the rough case we have seen.

General rough paths

Paths in $T^N(\mathbb{R}^d)$: for $N \in \mathbb{N}$, we consider a path $T^N(\mathbb{R}^d)$ -valued, so that

$$\mathbf{X} : \Delta_{[0,T]} \longrightarrow T^N(\mathbb{R}^d)$$

$$(s,t) \longmapsto \mathbf{X}_{s,t} := \left(1, X_{s,t}^{(1)}, \dots, X_{s,t}^{(N)} \right)$$

where $X_{s,t}^{(j)} = \{ X_{s,t}^\omega : |\omega| = j, \omega \in (\mathbb{R}^d)^{\otimes j}, j=0, \dots, N \}$.
 $\uparrow$ now they are abstract objects

Seminorm on this space: we can define a seminorm on the space of path $T^N(\mathbb{R}^d)$ -valued:

$$\|\mathbf{X}\|_{(\alpha,N)} = \max_{j=1,\dots,N} \|X^{(j)}\|_{j\alpha}^{\frac{1}{j}}$$

where $\|X^{(j)}\|_{j\alpha}$ is the standard α -norm so that $\|X^{(j)}\|_{j\alpha} := \sup_{0 \leq s < t \leq T} \frac{|X^{(j)}|_{j\alpha}}{|t-s|^{j\alpha}}$

($|X^{(j)}|_{j\alpha}$ is the norm in the tensor product). Obviously the seminorm give us the quasi-distance

$$\|\mathbf{X} ; \tilde{\mathbf{X}}\|_{(\alpha,N)} = \max_{j=1,\dots,N} \|X^{(j)} - \tilde{X}^{(j)}\|_{j\alpha}^{\frac{1}{j}}$$

α -Hölder rough path: let $\alpha \in (0,1]$, $N := \lfloor \frac{1}{\alpha} \rfloor$, i.e. the integer $N \in \mathbb{N}$ s.t. $\alpha N \leq 1 < \alpha(N+1)$.

We say that $\mathbf{X}$, a $T^N(\mathbb{R}^d)$ -path, is an α -Hölder rough path over $\mathbb{R}^d$ if

$$\|\mathbf{X}\|_{(\alpha,N)} < \infty;$$

• it holds the Chen relation;

$$\bar{X}_{s,t} = \bar{X}_{s,u} * X_{u,t} \quad \forall 0 \leq s \leq u \leq t \leq T$$